An Efficient Near-Optimal Algorithm for Adversarial m-Set Bandits

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Abstract

We study adversarial combinatorial bandits with m-set actions, where at each round the learner selects m out of d items and observes only the aggregate loss of the selected items. The resulting action set contains $K = \binom{d}{m}$ elements and can therefore be exponentially large. Nevertheless, the loss of every action is determined by the same d-dimensional vector of item losses. We propose a computationally efficient algorithm that exploits this structure without explicitly enumerating the action set. Against adaptive non-anticipating adversaries, it guarantees, with probability at least $1-\delta$, regret against the best fixed action of

$$R_T = O\left(\sqrt{dT\log(K/\delta)}\right).$$

This matches the high-probability regret bound of the finite-action EXP3–KW algorithm of Zimmert and Lattimore [29, Theorem 6 and Algorithm 3], whose direct implementation may require exponential space. Our algorithm instead represents each sampling distribution with d parameters and runs in polynomial time without enumerating the action set. Thus, it resolves the open problem posed by Maiti et al. [26].

1 Introduction

In this paper, we study adversarial m-set bandits. At each round, the learner selects m out of d items and incurs the sum of their losses, but observes only the aggregate loss, not the loss of each selected item. Since any m-subset may be chosen, the learner has $K = \binom{d}{m}$ possible actions, and its goal is to minimize regret with respect to the best fixed m-set in hindsight.

For finite action sets, Zimmert and Lattimore [29, Theorem 6 and Algorithm 3] show that EXP3–KW achieves a high-probability regret bound that, for m-sets, becomes $\widetilde{O}(\sqrt{dmT})$ when $m \le d/2$.[1] However, its direct implementation maintains one weight for every action, and since $K = \binom{d}{m}$, this can require exponential space. A possible workaround is to use the results in [26]. Specifically, Maiti et al. [26, Appendix E.6] give a polynomial-time high-probability algorithm based on a general directed acyclic graph representation that, for m-sets, turns into an algorithm achieving $\widetilde{O}(d\sqrt{mT})$ regret bound. Despite this implementation being computationally efficient, it remains unclear whether one can efficiently obtain the sharper rate $\widetilde{O}(\sqrt{dmT})$.

We provide a positive answer to this question with a polynomial-time algorithm. Against adaptive non-anticipating adversaries, our algorithm guarantees $\widetilde{O}(\sqrt{dsT})$ regret with high probability,

[1]The notation $\widetilde{O}$ suppresses logarithmic factors.

where $s := \min\{m, d-m\}$. In the regime $m \le d/2$, we have $s = m$, and the bound becomes $\widetilde{O}(\sqrt{dmT})$, and thus it matches the regret rate of finite-action EXP3–KW by Zimmert and Lattimore [29, Theorem 6 and Algorithm 3].

To obtain our efficient algorithm, we work with weighted m-set distributions, a family of distribution that can be represented by d item weights, from which one can sample and compute the required moments in polynomial time [11, 24]. The high-level idea is to implement the finite-action EXP3–KW algorithm while keeping the sampling distributions at all iterates within this family. The main obstacle is that the updates of EXP3-KW come with a quadratic correction, which does not preserve the weighted m-set structure. We overcome this difficulty by using the covariance bound of Cesari and Colomboni [10, Corollary 1.3] to construct a carefully chosen affine majorant of the quadratic correction on the m-set action space: exponential-weights updates using this majorant preserve the family of weighted m-set distributions, while the majorant also provides the estimation-error control needed to obtain the desired high-probability regret rate.

2 Setting and problem formulation

Preliminary definitions. For a positive integer n, we write $[n] := \{1, \dots, n\}$. For a vector $z = (z_1, \dots, z_n)$ and $k \in [n]$, we define its k-th elementary symmetric polynomial as

$$e_k(z) := \sum_{S \subseteq [n] :\, |S| = k} \prod_{i \in S} z_i, \qquad e_0(z) := 1, \qquad e_k(z) := 0 \quad \text{for } k < 0 \text{ or } k > n.$$

Fix two integers $d \ge 2$ and $1 \le m \le d-1$. We define the action set and its size as

$$\mathcal{X} := \mathcal{X}_{d,m} := \left\{ x \in \{0,1\}^d : \|x\|_1 = m \right\}, \qquad K := |\mathcal{X}| = \binom{d}{m}. \tag{1}$$

Each action $x \in \mathcal{X}$ selects exactly m of the d items, $x_i = 1$ if item i is selected and $x_i = 0$ otherwise. Its support is $\operatorname{supp}(x) := \{i \in [d] : x_i = 1\}$. Thus, the support of each action is an m-element subset of $[d]$, which we call an *m-set*. Conversely, each m-set $S \subseteq [d]$ corresponds to the indicator vector $\mathbf{1}_S \in \mathcal{X}$. We therefore use actions in $\mathcal{X}$ and m-sets interchangeably.

Let $s := \min\{m, d-m\}$. By the symmetry of the binomial coefficient, $\log K \le s \log(ed/s)$.

Linear-algebra notation. For a symmetric matrix $A \in \mathbb{R}^{d \times d}$, we denote its kernel by $\ker(A) := \{z \in \mathbb{R}^d : Az = 0\}$ and its Moore–Penrose pseudoinverse by A^+. For two symmetric matrices $A, B \in \mathbb{R}^{d \times d}$, we write $A \succeq B$ if $A - B$ is positive semidefinite or, equivalently

$$z^\top A z \ge z^\top B z \qquad \text{for every } z \in \mathbb{R}^d.$$

We write $A \succ 0$ if $z^\top A z > 0$ for every nonzero $z \in \mathbb{R}^d$. In particular, A is positive semidefinite if $A \succeq 0$, and positive definite if $A \succ 0$. Let $\mathbf{1} := (1, \dots, 1)^\top$, and define

$$\operatorname{span}\{\mathbf{1}\} := \{a\mathbf{1} : a \in \mathbb{R}\}, \qquad H_0 := \{y \in \mathbb{R}^d : \langle y, \mathbf{1} \rangle = 0\}.$$

Thus, H_0 is the subspace orthogonal to $\operatorname{span}\{\mathbf{1}\}$. If A is positive semidefinite and $\ker(A) = \operatorname{span}\{\mathbf{1}\}$, then A is invertible on H_0. For every $y \in H_0$, the vector $A^+ y$ is the inverse of A applied to y within this subspace.

We will use the order-reversing property of matrix inversion [18, Corollary 7.7.4(a)]. Specifically, suppose that A and B are positive semidefinite linear operator with

$$\ker(A) = \ker(B) = \operatorname{span}\{\mathbf{1}\},$$

then their restrictions to $H_0 = \mathbf{1}^\perp$ are positive definite. Consequently, if $A \succeq cB$, for $c > 0$, then

$$(A|_{H_0})^{-1} \preceq \frac{1}{c} (B|_{H_0})^{-1}.$$

Equivalently,

$$A^+ \preceq \frac{1}{c} B^+.$$

In particular,

$$y^\top A^+ y \le \frac{1}{c} y^\top B^+ y \qquad \text{for every } y \in H_0.$$

Probability notation. For a finite set $\mathcal{Y} \subseteq \mathbb{R}^d$, we denote its probability simplex by

$$\Delta(\mathcal{Y}) := \left\{ p : \mathcal{Y} \to [0,1] : \sum_{y \in \mathcal{Y}} p(y) = 1 \right\}.$$

For $p, q \in \Delta(\mathcal{Y})$, we define the entropy of p and the Kullback–Leibler (KL) divergence from p to q as

$$H(p) := -\sum_{y \in \mathcal{Y}} p(y) \log p(y), \qquad \mathrm{KL}(p \,\|\, q) := \sum_{y \in \mathcal{Y}} p(y) \log \frac{p(y)}{q(y)}.$$

We use the convention $0 \log(0/q) = 0$, and set $\mathrm{KL}(p \,\|\, q) := +\infty$ if $p(y) > 0 = q(y)$ for some $y \in \mathcal{Y}$.

Let X be a random vector drawn from $p \in \Delta(\mathcal{Y})$. We define the marginal vector, second-moment matrix, and covariance matrix of p as

$$\mu(p) := \mathbb{E}_{X \sim p}[X], \qquad M(p) := \mathbb{E}_{X \sim p}[XX^\top], \qquad \Sigma(p) := \mathrm{Cov}_{X \sim p}(X) = M(p) - \mu(p)\mu(p)^\top.$$

When $\mathcal{Y} = \mathcal{X}$, the i-th marginal probability is

$$\mu_i(p) = \mathbb{E}_{X \sim p}[X_i] = \mathbb{P}_{X \sim p}[X_i = 1].$$

Thus, $\mu_i(p)$ is the probability that action X selects item i. If $M(p)$ is invertible, we refer to the inverse-design quadratic form

$$x^\top M(p)^{-1} x$$

as the *leverage score* of x under p.

Weighted m-set distributions. Given $\theta \in \mathbb{R}^d$, define $w_i := e^{\theta_i} > 0$ for every $i \in [d]$. For every m-set $S \subseteq [d]$, the weighted m-set distribution p_θ is

$$p_\theta(S) := \frac{\exp\left(\sum_{i \in S} \theta_i\right)}{\sum_{A \subseteq [d] : |A| = m} \exp\left(\sum_{i \in A} \theta_i\right)} = \frac{\prod_{i \in S} w_i}{e_m(w)}.$$

Its log-partition function is

$$F(\theta) := \log \sum_{A \subseteq [d] : |A| = m} \exp\left(\sum_{i \in A} \theta_i\right) = \log e_m(e^{\theta_1}, \ldots, e^{\theta_d}).$$

If $S \sim p_\theta$ and $X := \mathbf{1}_S \in \mathcal{X}$ is its indicator vector, then

$$\nabla F(\theta) = \mathbb{E}_{X \sim p_\theta}[X] = \mu(p_\theta), \qquad \nabla^2 F(\theta) = \mathrm{Cov}_{X \sim p_\theta}(X) = \Sigma(p_\theta).$$

Although p_θ may assign positive probability to exponentially many actions, the d parameters $\theta_1, \ldots, \theta_d$ represent it. Finally, we define a set of distributions whose marginal probabilities stay away from 0 and 1. Let $r := m/d$ and fix $\lambda \in (0, 1)$. Define

$$\mathcal{P}_\lambda := \{p \in \Delta(\mathcal{X}) : \lambda r \leq \mu_i(p) \leq 1 - \lambda(1 - r) \text{ for every } i \in [d]\}\,. \tag{2}$$

The uniform distribution U over $\mathcal{X}$ satisfies $\mu_i(U) = r$ for every $i \in [d]$. Therefore, $U \in \mathcal{P}_\lambda$.

Adversarial m-set bandit protocol. Fix a time horizon $T \geq 1$. Let $(\mathcal{F}_t)_{t=0}^T$ denote the filtration generated by the interaction history, where $\mathcal{F}_{t-1}$ contains the information available before round t. At each round $t \in [T]$, the learner selects an $\mathcal{F}_{t-1}$-measurable distribution $p_t \in \Delta(\mathcal{X})$, where $\mathcal{X}$, defined in Equation (1), represents the set of actions available to the learner. The adversary selects a loss vector $\ell_t \in \mathbb{R}^d$ that is also $\mathcal{F}_{t-1}$-measurable. Thus, the adversary is *adaptive non-anticipating*. The loss vector ℓ_t may depend on the entire interaction history up to round $t - 1$, but not on the learner's random draw at round t. Therefore, conditionally on $\mathcal{F}_{t-1}$, both p_t and ℓ_t are fixed. The learner then draws $X_t \sim p_t$ and observes only the loss $Y_t := \langle X_t, \ell_t \rangle$. We use the standard action-level normalization

$$|\langle x, \ell_t \rangle| \leq 1 \qquad \forall x \in \mathcal{X},\ t \in [T],$$

also adopted by Maiti et al. [26, Assumption 1] and Zimmert and Lattimore [29, Section 1]. [2] The learner's realized regret after T rounds, measured against the best fixed action in hindsight, is

$$R_T := \sum_{t=1}^T \langle X_t, \ell_t \rangle - \min_{x \in \mathcal{X}} \sum_{t=1}^T \langle x, \ell_t \rangle.$$

Throughout, we use the shorthand $\mathbb{E}_t[\,\cdot\,] := \mathbb{E}[\,\cdot\,|\mathcal{F}_{t-1}]$ for conditional expectation given the information available before round $t \in [T]$.

3 Original contributions

Our main contribution is an algorithm (Algorithm 1) for adversarial m-set bandits that achieves the regret guarantee summarized in the following.

Theorem (Informal version of Theorem 1)**.** *There exists an algorithm that can be implemented in polynomial time such that, against any adaptive non-anticipating adversary, with probability at least $1 - \delta$, it achieves*

$$R_T = O\left(\sqrt{dT\left(\log\binom{d}{m} + \log\frac{1}{\delta}\right)}\right).$$

For example, one may take Algorithm 1.

We first discuss the state-of-the-art results for adversarial m-set bandits. Maiti et al. [26, Appendix E.6] obtain a high-probability regret bound of $\widetilde{O}(d\sqrt{mT})$ for m-set bandits and leave open whether the extra $\sqrt{d}$ factor in this bound can be removed with an efficient algorithm. Although this consequence is not stated explicitly in their paper, applying their general DAG guarantee [26, Theorem 7] to the m-set DAG constructed in Maiti et al. [26, Appendix E.6], which has $O(md)$

[2] More generally, if the action losses are bounded in absolute value by L, rescaling gives the same result with the regret bound multiplied by L.

edges and $K = \binom{d}{m}$ paths, and using $\log K \le m \log(ed/m)$ yields, for $m \le d/2$, the regret bound $\widetilde{O}(m\sqrt{dT})$.

Our algorithm achieves a regret upper bound of $\widetilde{O}(\sqrt{dsT})$, where $s := \min\{m, d-m\}$. In particular, when $m \le d/2$, we have $s = m$, and the bound becomes $\widetilde{O}(\sqrt{dmT})$. Thus, our bound improves the $\widetilde{O}(d\sqrt{mT})$ guarantee of Maiti et al. [26, Appendix E.6] by a factor of $\sqrt{d}$, and the derived $\widetilde{O}(m\sqrt{dT})$ consequence above [26, Theorem 7 and Appendix E.6] by a factor of $\sqrt{m}$. Consequently, relative to the better of these two bounds in the regime $m \le d/2$, the improvement is a factor of $\sqrt{m}$.

Zimmert and Lattimore [29, Theorem 6 and Algorithm 3] obtain a high-probability regret bound that, when $m \le d/2$, yields the rate $\widetilde{O}(\sqrt{dmT})$, matching our regret bound. However, their finite-action exponential-weights algorithm is stated with one weight for each of the $K = \binom{d}{m}$ actions, which can be exponential in d. Our algorithm achieves the same regret rate using only d parameters and running in polynomial time (see Section 5.3 for the implementation details).

In Table 1, we summarize the best known high-probability guarantees with $\sqrt{T}$ dependence for adversarial m-set bandits. All results reported there consider action-normalized losses and adaptive non-anticipating adversaries.

We next discuss the regret lower bound in the regime $m \le d/2$, making its dependence on $d-m$ explicit. Let $\rho := (d-m)/d$. When the loss vectors belong to $\{0,1\}^d$ Ito et al. [19, Theorem 2] prove the expected-regret lower bound

$$\Omega\left(\min\left\{\rho^2\sqrt{dm^3T}, \rho m^{3/4} T\right\}\right).$$

To compare this result with our action-normalized setting, observe that an m-set may incur a loss as large as m when the loss vectors belong to $\{0,1\}^d$. We therefore divide the losses by m. Since regret scales linearly with the losses, this yields the action-normalized lower bound

$$\Omega\left(\min\left\{\rho^2\sqrt{dmT}, \frac{\rho T}{m^{1/4}}\right\}\right).$$

Consequently, when $m \le d/2$ and $T \ge dm^{3/2}$, this lower bound is $\Omega(\sqrt{dmT})$. On the other hand, setting $\delta = T^{-2}$ in our high-probability guarantee and using $R_T \le 2T$ yields an expected-regret upper bound of $\widetilde{O}(\sqrt{dmT})$. Therefore, our rate matches the minimax expected-regret lower bound up to logarithmic factors in this regime. In particular, any improvement in the polynomial dependence of a uniform high-probability bound valid for $\delta = T^{-2}$ would imply the same improvement in expected regret and would therefore contradict this lower bound.

Work	Regret bound	Computational complexity
Zimmert and Lattimore [29] (Theorem 6 and Algorithm 3)	$\widetilde{O}(\sqrt{dmT})$	Exponential
Maiti et al. [26] (Appendix E.6)	$\widetilde{O}(d\sqrt{mT})$	Polynomial
Maiti et al. [26] (Theorem 7 applied here to the DAG of Appendix E.6)	$\widetilde{O}(m\sqrt{dT})$	Polynomial
This work	$\boldsymbol{\widetilde{O}(\sqrt{dmT})}$	**Polynomial**

Table 1: Statistical and computational comparison for the regime $m \le d/2$. All upper bounds hold with high probability against adaptive non-anticipating adversaries.

Challenges and techniques. In their finite-action EXP3–KW algorithm, Zimmert and Lattimore [29, Algorithm 3] obtain a high-probability regret bound using the biased surrogate loss

$$\langle x, \widehat{\ell}_t \rangle - \eta x^\top M_t^{-1} x,$$

where $\eta > 0$ is the learning rate,

$$M_t = \mathbb{E}_{X \sim p_t}[XX^\top]$$

is the second-moment matrix of the sampling distribution p_t, and

$$\widehat{\ell}_t := M_t^{-1} X_t \langle X_t, \ell_t \rangle$$

is the loss estimate constructed after drawing $X_t \sim p_t$ and observing the scalar loss $\langle X_t, \ell_t \rangle$.

The main obstacle to an efficient implementation of this approach is that the leverage score is quadratic in x. Even if we start from a weighted m-set distribution and we insert this term directly into an action-level multiplicative-weights update, the updated distribution need not remain in the weighted m-set family and, consequently, it may require one weight for each of the $K = \binom{d}{m}$ actions.

Our first step is to replace the leverage score with an affine upper bound. Starting from the covariance inequality of Cesari and Colomboni [10, Corollary 1.3], we derive

$$x^\top M_t^{-1} x \le \phi_t(x) := 1 + 2 \sum_{i=1}^{d} \frac{(x_i - \mu_{t,i})^2}{\mu_{t,i}(1 - \mu_{t,i})}. \tag{3}$$

Although the expression defining ϕ_t appears quadratic, it is affine on $\mathcal{X}$. Indeed, every $x \in \mathcal{X}$ satisfies $x_i^2 = x_i$ and $\sum_{i=1}^d x_i = m$. For later convenience, we scale this affine majorant by four and write

$$c_t^\top x = 4\phi_t(x) \qquad \text{for every } x \in \mathcal{X}.$$

We then define the surrogate loss

$$Z_t(x) := \langle x, \widehat{\ell}_t \rangle - \eta c_t^\top x. \tag{4}$$

The correction in Equation (4) plays the same role as the quadratic correction used by Zimmert and Lattimore [29, Algorithm 3], but remains affine in x. Consequently, the multiplicative-weights step maps a weighted m-set distribution to another weighted m-set distribution.

A further difficulty is that the affine leverage majorant can become arbitrarily large when a marginal probability approaches 0 or 1. Consequently, the estimated action losses $\langle x, \widehat{\ell}_t \rangle$ can have arbitrarily large one-step magnitude and conditional second moment, preventing both the second-order exponential-weights analysis and the required high-probability concentration arguments. To obtain the uniform leverage bound needed in the analysis, we project the updates toward the set $\mathcal{P}_\lambda$ introduced in Section 2.

More precisely, the exact constrained OMD iterate is the KL projection of the unprojected exponential-weights distribution $\widetilde{p}_{t+1}$ onto $\mathcal{P}_\lambda$. Our polynomial-time approximation is instead guaranteed to belong to $\mathcal{P}_{\lambda/2}$, where the extra slack serves to accommodate the finite accuracy of the projection procedure. The approximate KL update also preserves the weighted m-set family, as shown in Lemma 11. Hence, every iterate can be represented as p_{θ_t} using the d parameters

$$\theta_{t,1}, \dots, \theta_{t,d},$$

rather than one weight for each of the K actions.

The comparator in the OMD analysis must also belong to $\mathcal{P}_\lambda$. However, the distribution δ_x, which puts all its probability on x, has marginal probabilities in $\{0, 1\}$ and does not belong to $\mathcal{P}_\lambda$. We therefore use the smoothed comparator

$$q_x := (1 - \lambda)\delta_x + \lambda U, \tag{5}$$

where U is the uniform distribution over $\mathcal{X}$. By construction, $q_x \in \mathcal{P}_\lambda$. Since the losses are action-normalized, using q_x in place of δ_x adds at most $2\lambda T$ to the regret.

The remaining analytical challenge is to control the estimation error under the smoothed comparator q_x, namely,

$$\sum_{t=1}^{T} \left(\mathbb{E}_{X \sim q_x}[\langle X, \widehat{\ell}_t \rangle] - \mathbb{E}_{X \sim q_x}[\langle X, \ell_t \rangle] \right).$$

Applying a concentration bound to this estimation error introduces the positive term

$$\frac{\eta}{4} \sum_{t=1}^{T} \mathbb{E}_{X \sim q_x}[c_t^\top X].$$

Controlling this term separately would not give the desired regret bound. However, the affine correction in Equation (4) contributes

$$-\eta \sum_{t=1}^{T} \mathbb{E}_{X \sim q_x}[c_t^\top X]$$

to the OMD bound. Since $c_t^\top X = 4\phi_t(X) \geq 0$, this negative contribution dominates the positive concentration term. Their sum is therefore non-positive and may be dropped from the regret upper bound.

Finally, the elementary-symmetric-polynomial recurrence of Chen and Liu [11, Section 2, Method 2] allows us to compute the marginal probabilities and second-moment matrix of p_t in polynomial time. The associated conditional-Bernoulli procedure gives an exact sample from p_t without enumerating $\mathcal{X}$; see Chen and Liu [11, Section 4, Procedure 3], as well as the alternative backward-path construction in Procedure 5. This sampling rule is also the $L = \mathrm{diag}(w)$ specialization of Kulesza and Taskar [24, Section 3.1 and Algorithm 2]. Together with the fact that both the affine update and the approximate KL update preserve the family of weighted m-set distributions, these routines establish the polynomial-time implementation in Proposition 1.

4 Related work

Optimal design and finite-action linear bandits. Kiefer [21] introduced the general framework of approximate optimal design, and Kiefer and Wolfowitz [22] proved the equivalence between D- and G-optimality. For a finite action set $\mathcal{A} \subset \mathbb{R}^d$, Bubeck et al. [8, Theorem 4] combined exponential weights with John's exploration and obtained $O(\sqrt{dT \log |\mathcal{A}|})$ expected regret under action-normalized losses. For general convex action sets, Abernethy et al. [2, Theorem 1] used self-concordant barriers to obtain an efficient expected-regret bound of $\widetilde{O}(d^{3/2}\sqrt{T})$.

The first general high-probability bound with $\sqrt{T}$ dependence against adaptive adversaries had a larger dependence on the dimension. Bartlett et al. [5, Theorem 1] obtained $\widetilde{O}(d^{3/2}\sqrt{T})$ regret. Abernethy and Rakhlin [1, Theorem 4 and Section 5.4] then gave a general high-probability reduction. Its efficient implementation requires affine confidence bounds tailored to the geometry of the action set. Later, Lee et al. [25, Theorem 3.1] gave the first general polynomial-time high-probability algorithm, with worst-case regret $\widetilde{O}(d^{7/2}\sqrt{T})$. Zimmert and Lattimore [29, Theorem 4] improved this rate to $\widetilde{O}(d^2\sqrt{T})$.

For a finite action set $\mathcal{A}$, Zimmert and Lattimore [29, Theorem 6 and Algorithm 3] also combined EXP3 with Kiefer–Wolfowitz exploration and obtained $O\left(\sqrt{dT\log(|\mathcal{A}|/\delta)}\right)$ regret with high probability. We use this regret bound as our statistical benchmark. When applied to our action set $\mathcal{X} = \mathcal{X}_{d,m}$, a direct implementation maintains $K = \binom{d}{m}$ weights. However, K can be exponential in d.

Combinatorial optimization with bandit feedback. Early adversarial bandit algorithms for structured action sets were given by McMahan and Blum [27] and Awerbuch and Kleinberg [4]. Cesa-Bianchi and Lugosi [9] subsequently introduced ComBand, a general framework for adversarial combinatorial bandits. This line of work was subsequently developed by Audibert et al. [3, Section 4, after Theorem 5], who considered full-information, semi-bandit, and full-bandit feedback and established several lower bounds. In particular, they conjectured that the minimax regret in the bandit setting should scale as $m\sqrt{dT}$, where each action selects m out of d items and T is the time horizon. On the computational side, Combes et al. [14, Theorem 6 and 7] later introduced CombEXP, improving the computational complexity for some combinatorial action classes while preserving the existing regret scaling.

Cohen et al. [13, Theorems 1 and 5] disproved the conjecture of Audibert et al. [3]. Using correlated losses, they proved an expected-regret lower bound of $\Omega(\sqrt{dm^3T/\log T})$ for a multitask problem in which every action selects m items. Ito et al. [19, Theorem 2] later removed the $1/\sqrt{\log T}$ factor using binary correlated losses. For m-set bandits, their result also describes the dependence on $d - m$ and on the time horizon, as discussed in Section 3.

Efficient algorithms for structured action sets. Several works avoid enumerating exponentially large action sets by using a compact representation or an optimization oracle. For expected or pseudo-regret, Hazan and Karnin [17, Corollary 35] use volumetric spanners, while Ito et al. [20, Theorem 1] use an optimization oracle. Both approaches run in polynomial time and achieve $\widetilde{O}(\sqrt{T})$ regret with polynomial dependence on the dimension.

Earlier efficient high-probability guarantees with $T^{2/3}$ dependence were obtained by Braun and Pokutta [7, Theorem 3.1], using a linear-optimization oracle over the action polytope, and by Sakaue et al. [28, Theorem 1], using a zero-suppressed binary decision diagram. These approaches are efficient when the required oracle or compact representation is available, but do not attain the $\sqrt{T}$ dependence achieved here.

The work most closely related to ours is Maiti et al. [26, Appendix E.6]. They represent the actions as paths in a DAG and obtain an efficient algorithm with a high-probability regret guarantee against adaptive adversaries. For m-sets, their algorithm achieves $\widetilde{O}(d\sqrt{mT})$ regret, and they ask whether $\widetilde{O}(\sqrt{dmT})$ can be achieved efficiently. Our result answers this question with a polynomial-time algorithm.

Finally, Kontogiannis et al. [23, Theorem 3.2 and Appendix G.5] show that their kernelized payoff-based framework applies to m-sets and yields $\widetilde{O}(d^{2/3}m^{4/3}T^{2/3})$ high-probability regret. Our result improves the dependence on the horizon to $\sqrt{T}$.

Weighted m-set distributions. The weighted m-set distribution introduced in Section 2 is also known as a conditional Bernoulli or rejective-sampling distribution [11, 12, 16]. Classical methods based on elementary symmetric polynomials allow us to compute its normalizing constant and marginal probabilities, and to sample from it efficiently [11, 24].

Beyond these computational properties, weighted m-set distributions have a covariance structure central to our analysis: Cesari and Colomboni [10, Corollary 1.3] prove exactly the half-normalized covariance bound used below.

5 A near-optimal algorithm for m-set bandits

5.1 Affine upper bound on the leverage score

In this section, we derive the affine upper bound on the leverage score $x^\top M^{-1}x$ used by our algorithm. Complete proofs of the consequences derived here are in Section A. The covariance bound itself is imported from Cesari and Colomboni [10, Corollary 1.3] and is not reproved here.

Lemma 1 (Cesari–Colomboni covariance bound)**.** *Let $p \in \Delta(\mathcal{X})$ be a weighted m-set distribution with positive weights, where $1 \le m \le d-1$, and let Σ be its covariance matrix. For every $i \in [d]$, define $v_i := \Sigma_{ii}$. Then*

$$\Sigma \succeq \frac{1}{2}\left(D - \frac{vv^\top}{V}\right),$$

where $v := (v_1, \ldots, v_d)^\top$, $D := \operatorname{diag}(v)$, and $V := \sum_{i=1}^d v_i$.

Using Lemma 1, we prove the following bound on the Moore–Penrose pseudoinverse Σ^+ of Σ.

Lemma 2. *Let $p \in \Delta(\mathcal{X})$ be a weighted m-set distribution with positive weights, where $1 \le m \le d-1$. Let Σ be its covariance matrix, and define $v_i := \Sigma_{ii}$ for every $i \in [d]$. Then,* $\ker(\Sigma) = \operatorname{span}\{\mathbf{1}\}$, *and, for every $y \in H_0 := \{z \in \mathbb{R}^d : \langle z, \mathbf{1}\rangle = 0\}$, we have*

$$y^\top \Sigma^+ y \le 2\sum_{i=1}^d \frac{y_i^2}{v_i}.$$

Proof sketch. Let $P := D - vv^\top/V$. A direct calculation, given in Section A.1, shows that $\ker(P) = \ker(\Sigma) = \operatorname{span}\{\mathbf{1}\}$. Therefore, both matrices are positive definite, and hence invertible, on H_0. On H_0, the bound in Lemma 1 gives

$$\Sigma \succeq \frac{1}{2}P.$$

Thus, on H_0, we have

$$y^\top \Sigma^+ y \le 2y^\top P^+ y \qquad \text{for every } y \in H_0. \tag{6}$$

We now relate P^+ to D^{-1}. Since $y \in H_0$, we have $\sum_{i=1}^d y_i = 0$. Therefore,

$$PD^{-1}y = \left(D - \frac{vv^\top}{V}\right)D^{-1}y = y - \frac{v}{V}\sum_{i=1}^d y_i = y. \tag{7}$$

We also have

$$P(P^+y) = y, \tag{8}$$

because P^+ coincides with the inverse of P on H_0. Subtracting Equation (8) from Equation (7) gives $P(D^{-1}y - P^+y) = 0$. It follows that

$$D^{-1}y - P^+y \in \ker(P) = \operatorname{span}\{\mathbf{1}\}.$$

Since $y \in H_0$, this difference is orthogonal to y. Thus,

$$y^\top P^+ y = y^\top D^{-1} y = \sum_{i=1}^{d} \frac{y_i^2}{v_i}. \tag{9}$$

Combining (6) and (9) proves the result. See Section A.1 for the complete proof. □

The next lemma relates the pseudoinverse Σ^+ of the covariance matrix of a weighted m-set distribution to the inverse M^{-1} of its second-moment matrix.

Lemma 3. *Let $p \in \Delta(\mathcal{X})$ be a weighted m-set distribution with positive weights and $1 \le m \le d-1$. Then its second-moment matrix M is positive definite and, for every $y \in H_0$,*

$$y^\top M^{-1} y = y^\top \Sigma^+ y.$$

Consequently, for every $x \in \mathcal{X}$,

$$x^\top M^{-1} x = 1 + (x-\mu)^\top \Sigma^+ (x-\mu),$$

where $\mu = \mu(p)$ is the marginal vector of p.

Proof sketch. If $a^\top M a = 0$, full support implies $a^\top x = 0$ for every $x \in \mathcal{X}$. Comparing two actions that differ by swapping items i and j forces $a_i = a_j$. Hence $a = c\mathbf{1}$, and then $0 = a^\top x = cm$ gives $c = 0$. Thus $M \succ 0$. Hence, $M\mathbf{1} = \mathbb{E}_{X \sim p}[XX^\top \mathbf{1}] = m\mu$. In addition, since M is invertible,

$$M^{-1}\mu = \frac{\mathbf{1}}{m}. \tag{10}$$

Now fix $y \in H_0$. By Equation (10),

$$\mu^\top M^{-1} y = (M^{-1}\mu)^\top y = \frac{1}{m}\mathbf{1}^\top y = 0.$$

Therefore,

$$\Sigma M^{-1} y = (M - \mu\mu^\top) M^{-1} y = y.$$

Since Σ^+ is the inverse of Σ on H_0, the vectors $M^{-1}y$ and $\Sigma^+ y$ differ by an element of $\ker(\Sigma) = \operatorname{span}\{\mathbf{1}\}$. This difference is orthogonal to y, and thus

$$y^\top M^{-1} y = y^\top \Sigma^+ y. \tag{11}$$

Finally, fix $x \in \mathcal{X}$. Since both x and μ have coordinates that sum to m, we have $x - \mu \in H_0$. Expanding $x = \mu + (x-\mu)$ and using Equation (10) and Equation (11) gives

$$x^\top M^{-1} x = 1 + (x-\mu)^\top \Sigma^+ (x-\mu).$$

See Section A.2 for the complete proof. □

We now combine Lemmas 2 and 3 to obtain the affine upper bound on the leverage score used by our algorithm.

Lemma 4. *Let $p \in \mathcal{P}_\lambda$ be a weighted m-set distribution, where $1 \le m \le d-1$, and let μ and M be its marginal vector and second-moment matrix. For every $x \in \mathcal{X}$, define*

$$\phi_p(x) := 1 + 2\sum_{i=1}^{d} \frac{(x_i - \mu_i)^2}{\mu_i(1-\mu_i)}.$$

Then, for every $x \in \mathcal{X}$, $x^\top M^{-1} x \le \phi_p(x)$. Moreover, $4\phi_p(x) = c_p^\top x$, where

$$c_{p,i} = 4\left(\frac{1 + 2\sum_{j=1}^{d} \frac{\mu_j}{1-\mu_j}}{m} + 2\frac{1-2\mu_i}{\mu_i(1-\mu_i)}\right) \qquad \textit{for every } i \in [d].$$

Finally, $\mathbb{E}_{X\sim p}[\phi_p(X)] = 2d+1$, and, for every $x \in \mathcal{X}$, $0 \le \phi_p(x) \le 1 + 4d/\lambda$.

Proof sketch. Fix $x \in \mathcal{X}$. Since both x and μ have coordinates that sum to m, we have $x - \mu \in H_0$. Therefore, Lemmas 2 and 3 give

$$x^\top M^{-1} x = 1 + (x-\mu)^\top \Sigma^+ (x-\mu) \le 1 + 2\sum_{i=1}^{d} \frac{(x_i-\mu_i)^2}{\mu_i(1-\mu_i)} = \phi_p(x).$$

Since $\mathbb{E}_{X\sim p}[(X_i - \mu_i)^2] = \mu_i(1-\mu_i)$, we have $\mathbb{E}_{X\sim p}[\phi_p(X)] = 1 + 2d$. Moreover, $p \in \mathcal{P}_\lambda$ implies $\mu_i \ge \lambda r$ and $1 - \mu_i \ge \lambda(1-r)$, where $r = m/d$. Separating the m coordinates for which $x_i = 1$ from the $d - m$ coordinates for which $x_i = 0$ gives

$$\sum_{i=1}^{d} \frac{(x_i-\mu_i)^2}{\mu_i(1-\mu_i)} \le \frac{m}{\lambda r} + \frac{d-m}{\lambda(1-r)} = \frac{2d}{\lambda}.$$

Hence,

$$0 \le \phi_p(x) \le 1 + \frac{4d}{\lambda}.$$

Finally, using $x_i^2 = x_i$, we obtain

$$\phi_p(x) = 1 + 2\sum_{j=1}^{d} \frac{\mu_j}{1-\mu_j} + 2\sum_{i=1}^{d} \frac{1-2\mu_i}{\mu_i(1-\mu_i)} x_i.$$

Since $\sum_i x_i = m$, we can write the constant term as a linear function of x. Multiplying the resulting affine representation by four gives $4\phi_p(x) = c_p^\top x$, with c_p defined in the statement. See Section A.3 for the complete proof. $\square$

5.2 Algorithm and high-probability regret

In this section, we present Algorithm 1 and prove its high-probability regret guarantee. At each round t, the learner draws an action X_t from a weighted m-set distribution $p_t \in \mathcal{P}_{\lambda/2}$ and observes only the scalar loss $\langle X_t, \ell_t \rangle$. It uses this observation to construct the KW loss estimate $\widehat{\ell}_t$.

At round t, we set $\phi_t := \phi_{p_t}$ and $c_t := c_{p_t}$, where ϕ_p and c_p are defined in Lemma 4. The learner forms a biased surrogate loss by subtracting the affine correction $\eta c_t^\top x$ from $\langle x, \widehat{\ell}_t \rangle$. Finally, the learner computes an approximate entropic OMD update toward $\mathcal{P}_\lambda$ (defined in Eq. (2)). The update is accurate enough for the OMD analysis and returns a distribution in $\mathcal{P}_{\lambda/2}$, so its marginal probabilities stay away from 0 and 1. We state the resulting regret bound in Theorem 1.

Algorithm 1 Affine KW–OMD with approximate entropic updates toward $\mathcal{P}_\lambda$

Require: $T \geq 1, \delta \in (0,1)$

Set $\eta \leftarrow \min\left\{\frac{1}{256d}, \sqrt{\frac{\log(12K/\delta)}{320dT}}\right\}$

Set $\lambda \leftarrow 128\eta d$ and $\varepsilon_{\mathrm{p}} \leftarrow \eta/T$

Initialize $\theta_1 \leftarrow 0$ and $p_1 \leftarrow p_{\theta_1} = U$ ▷ U is a uniform distribution

for $t = 1, \ldots, T$ **do**

  Compute $\mu_t \leftarrow \mu(p_t)$, $M_t \leftarrow M(p_t) = \mathbb{E}_{X \sim p_t}[XX^\top]$

  Set $\phi_t \leftarrow \phi_{p_t}$ and $c_t \leftarrow c_{p_t}$, so that, for every $x \in \mathcal{X}$,

$$\phi_t(x) \leftarrow 1 + 2\sum_{i=1}^{d} \frac{(x_i - \mu_{t,i})^2}{\mu_{t,i}(1-\mu_{t,i})} \quad \text{and} \quad c_t^\top x = 4\phi_t(x)$$

  Draw $X_t \sim p_t$

  Observe $Y_t = \langle X_t, \ell_t \rangle$

  Compute $\widehat{\ell}_t \leftarrow M_t^{-1} X_t Y_t$

  Define $Z_t(x) \leftarrow \langle x, \widehat{\ell}_t \rangle - \eta c_t^\top x$, for every $x \in \mathcal{X}$

  Set $z_t \leftarrow \widehat{\ell}_t - \eta c_t$, $\widetilde{\theta}_{t+1} \leftarrow \theta_t - \eta z_t$, and $\widetilde{p}_{t+1} \leftarrow p_{\widetilde{\theta}_{t+1}}$

  Apply a fixed deterministic implementation of the procedure in Lemma 11 to obtain $p_{t+1} = p_{\theta_{t+1}}$ satisfying

$$p_{t+1} \in \mathcal{P}_{\lambda/2}$$

  and, simultaneously for every $q \in \mathcal{P}_\lambda$,

$$\mathrm{KL}(q \| p_{t+1}) + \mathrm{KL}(p_{t+1} \| \widetilde{p}_{t+1}) \leq \mathrm{KL}(q \| \widetilde{p}_{t+1}) + \varepsilon_{\mathrm{p}}.$$

end for

Theorem 1 (High-probability KW bound)**.** *There exists a universal constant $C > 0$ such that the following holds. Let $T \geq 1$ be an integer, let $\delta \in (0,1)$, and assume $1 \leq m \leq d-1$. Run Algorithm 1. Then, with probability at least $1 - \delta$,*

$$R_T \leq C\sqrt{dT\left(\log K + \log\frac{1}{\delta}\right)},$$

where $K = \binom{d}{m}$. One may take $C = 160$.

Proof sketch. For every $x \in \mathcal{X}$, define the smoothed comparator

$$q_x := (1-\lambda)\delta_x + \lambda U,$$

where δ_x puts all its probability on x and U is the uniform distribution over $\mathcal{X}$. By construction, $q_x \in \mathcal{P}_\lambda$. The concentration bounds below hold simultaneously for every $x \in \mathcal{X}$, so we can then choose x to be a best action.

By adding and subtracting the expected losses under p_t and q_x, and applying Lemma 9, we obtain, with high probability,

$$R_T \le \underbrace{\sum_{t=1}^{T} \left(\mathbb{E}_{X\sim p_t}[\langle X, \ell_t\rangle] - \mathbb{E}_{X\sim q_x}[\langle X, \ell_t\rangle] \right)}_{=:(\star)} + O\left(\sqrt{T \log \frac{1}{\delta}}\right) + O(\lambda T). \tag{12}$$

The square-root term controls the difference between the learner's realized and expected losses. The term $O(\lambda T)$ is the cost of using q_x in place of x. It remains to bound $(\star)$. Adding and subtracting the expectations of the estimated losses gives

$$\begin{aligned}(\star) &= \sum_{t=1}^{T} \left(\mathbb{E}_{X\sim p_t}[\langle X, \ell_t\rangle] - \mathbb{E}_{X\sim p_t}[\langle X, \widehat{\ell}_t\rangle] \right) \\ &\quad + \sum_{t=1}^{T} \left(\mathbb{E}_{X\sim p_t}[\langle X, \widehat{\ell}_t\rangle] - \mathbb{E}_{X\sim q_x}[\langle X, \widehat{\ell}_t\rangle] \right) + \sum_{t=1}^{T} \left(\mathbb{E}_{X\sim q_x}[\langle X, \widehat{\ell}_t\rangle] - \mathbb{E}_{X\sim q_x}[\langle X, \ell_t\rangle] \right). \end{aligned} \tag{13}$$

The first and third terms in (13) can be bounded by applying Lemma 7 under p_t and q_x, respectively. To bound the second term, recall the surrogate loss

$$Z_t(x) := \langle x, \widehat{\ell}_t\rangle - \eta c_t^\top x, \qquad x \in \mathcal{X}.$$

Since

$$\langle x, \widehat{\ell}_t\rangle = Z_t(x) + \eta c_t^\top x,$$

we can apply Lemma 6 and Lemma 8 to the surrogate losses Z_t. Moreover, Lemma 4 gives

$$\mathbb{E}_{X\sim p_t}[c_t^\top X] = 4(2d+1) = O(d).$$

Combining these results, we have

$$(\star) \le O\left(\frac{\log K + \log(1/\delta)}{\eta} + \eta d T + 1 \right) + \underbrace{\left(-\eta + \frac{\eta}{4} \right) \sum_{t=1}^{T} \mathbb{E}_{X\sim q_x}[c_t^\top X]}_{=:(\star\star)\le 0}. \tag{14}$$

The coefficient $-\eta$ in $(\star\star)$ comes from the affine correction $-\eta c_t^\top x$ in Z_t, while the coefficient $\eta/4$ comes from the concentration bound for the third term in (13). Since

$$c_t^\top X = 4\phi_t(X) \ge 0,$$

the term $(\star\star)$ is nonpositive. The additive constant in (14) is the accumulated error of the approximate KL updates: by the choice $\varepsilon_{\mathrm{p}} = \eta/T$, its total contribution is $T\varepsilon_{\mathrm{p}}/\eta = 1$. Substituting (14) into (12) gives

$$R_T \le O\left(\frac{\log K + \log(1/\delta)}{\eta} + \eta d T + 1 + \lambda T + \sqrt{T \log \frac{1}{\delta}} \right). \tag{15}$$

Finally, substituting $\lambda = 128\eta d$ and the value of η, and combining the resulting high-probability estimate with the deterministic bound $R_T \le 2T$, gives the stated result with $C = 160$. The complete two-case calculation is given in Section B.1. $\square$

Let $s := \min\{m, d-m\}$. Since $\log K \le s \log(ed/s)$, the bound in Theorem 1 is $\widetilde{O}(\sqrt{dsT})$. In particular, when $m \le d/2$, it is $\widetilde{O}(\sqrt{dmT})$.

5.3 Polynomial-time implementation

We now show that Algorithm 1 can be implemented in polynomial time without enumerating $\mathcal{X}$.

The key idea is to maintain p_t in the weighted m-set form p_{θ_t}, with $\theta_t \in \mathbb{R}^d$, thereby allowing all steps of the algorithm to be performed without enumerating $\mathcal{X}$. The convex problem used in the update is solved to the explicit inverse-polynomial objective accuracy established in Lemma 11.

Proposition 1. *Algorithm 1 can be implemented in* $T \cdot \mathrm{poly}(d, m, \log T)$ *time and* $\mathrm{poly}(d, m)$ *space, without enumerating* $\mathcal{X}$*. At every round,* p_t *is a weighted* m*-set distribution* p_{θ_t} *represented by the* d *parameters* $\theta_{t,1}, \dots, \theta_{t,d}$*.*

Proof sketch. We observe that if p_t has a weighted m-set representation, its moments can be computed and a sample can be drawn in polynomial time using the recurrences in Lemma 12.

The strategy update in Algorithm 1 can be decomposed into two steps. First, define the exponential update

$$\widetilde{p}_{t+1}(x) \propto p_t(x) \exp\big(-\eta Z_t(x)\big).$$

The exact update toward which we compute is the KL projection

$$p^\star_{t+1} \in \arg\min_{p \in \mathcal{P}_\lambda} \mathrm{KL}(p \| \widetilde{p}_{t+1}).$$

Since Z_t is affine, the exponential update only changes the d parameters, and $\widetilde{p}_{t+1}$ remains a weighted m-set distribution. The exact projection is characterized by a convex problem in $2d$ variables. Solving this problem to the objective accuracy in Lemma 11 returns a weighted m-set distribution $p_{t+1} \in \mathcal{P}_{\lambda/2}$ and introduces at most $\varepsilon_{\mathrm{p}} = \eta/T$ error in the KL inequality used by OMD. The required objective accuracy is inverse-polynomial, and the ellipsoid method obtains it using polynomially many evaluations of the objective and its gradient. Since the initial distribution p_1 is uniform and admits a weighted m-set representation, this representation is preserved at every round. Thus, all steps of the algorithm can be performed in polynomial time without enumerating $\mathcal{X}$. The complete proof is given in Section C.3. □

6 Conclusion and future work

We introduced an algorithm for adversarial m-set bandits whose high-probability regret matches the finite-action EXP3–KW rate up to logarithmic factors, while running in polynomial time without enumerating the action set.

Our results leave two main questions open. First, it remains open to determine whether the $\log K$ term in the leading bound can be replaced by

$$s := \min\{m, d - m\},$$

yielding $O(\sqrt{dsT})$ realized regret at constant confidence, or whether an additional logarithmic factor is unavoidable against adaptive non-anticipating adversaries. We remark, to the best of our knowledge, this issue is open already for ordinary adversarial multi-armed bandits, corresponding to $m = 1$. Second, it would be interesting to determine whether our approach extends beyond m-sets. A natural test case is that of matroid-base action sets, since m-sets are precisely the bases of a uniform matroid. Although affine exponential-weights updates still preserve weighted matroid-base distributions, our analysis also requires an efficiently computable affine leverage majorant with controlled expectation and range, as well as efficient moment and KL-projection routines. These properties do not follow from our present covariance-based construction, so extending the near-optimal high-probability guarantee to broader matroid classes would require additional structural and computational ideas.

Appendix

A Proofs from Section 5.1

A.1 Proof of Lemma 2

Lemma 2. *Let $p \in \Delta(\mathcal{X})$ be a weighted m-set distribution with positive weights, where $1 \le m \le d-1$. Let Σ be its covariance matrix, and define $v_i := \Sigma_{ii}$ for every $i \in [d]$. Then,* $\ker(\Sigma) = \operatorname{span}\{\mathbf{1}\}$, *and, for every $y \in H_0 := \{z \in \mathbb{R}^d : \langle z, \mathbf{1}\rangle = 0\}$, we have*

$$y^\top \Sigma^+ y \le 2\sum_{i=1}^d \frac{y_i^2}{v_i}.$$

Proof. Let

$$P := D - \frac{vv^\top}{V}.$$

We first identify the kernels of P and Σ. Since p has positive weights, it has full support on $\mathcal{X}$. Therefore,

$$z^\top \Sigma z = \operatorname{Var}_{X\sim p}(\langle z, X\rangle) = 0$$

only if $\langle z, x\rangle$ is the same for every $x \in \mathcal{X}$.

Fix $i \neq j$. Since $1 \le m \le d-1$, we can choose two actions that differ only by exchanging item i with item j. The equality of their inner products with z gives $z_i = z_j$. Thus, all the coordinates of z are equal. Conversely, $\langle \mathbf{1}, X\rangle = m$ for every $X \in \mathcal{X}$, so $\Sigma\mathbf{1} = 0$. Hence,

$$\ker(\Sigma) = \operatorname{span}\{\mathbf{1}\}. \tag{16}$$

Moreover, for every $z \in \mathbb{R}^d$,

$$z^\top P z = \sum_{i=1}^d v_i z_i^2 - \frac{\left(\sum_{i=1}^d v_i z_i\right)^2}{V} = \sum_{i=1}^d v_i (z_i - \bar{z}_v)^2,$$

where

$$\bar{z}_v := \frac{\sum_{i=1}^d v_i z_i}{V}.$$

Full support also gives

$$v_i = \operatorname{Var}(X_i) = \mu_i(1-\mu_i) > 0 \qquad \text{for every } i \in [d].$$

It follows that $z^\top P z = 0$ if and only if all the coordinates of z are equal. Therefore,

$$\ker(P) = \operatorname{span}\{\mathbf{1}\}. \tag{17}$$

By (16) and (17), the pseudoinverse order property stated in the preliminaries applies to the bound in Lemma 1. Hence, for every $y \in H_0$,

$$y^\top \Sigma^+ y \le 2 y^\top P^+ y. \tag{18}$$

It remains to compute $y^\top P^+ y$. Since $D = \operatorname{diag}(v)$, we have $v^\top D^{-1} = \mathbf{1}^\top$. Thus, for every $y \in H_0$,

$$PD^{-1}y = y - \frac{v}{V}\mathbf{1}^\top y = y. \tag{19}$$

Moreover, $\operatorname{range}(P) = H_0$, so

$$PP^+y = y. \tag{20}$$

Equations (19) and (20) show that $D^{-1}y$ and P^+y both solve the equation $Pz = y$. Therefore,

$$D^{-1}y - P^+y \in \ker(P) = \operatorname{span}\{\mathbf{1}\}.$$

Since $y \in H_0$, it is orthogonal to this difference. Consequently,

$$y^\top P^+y = y^\top D^{-1}y = \sum_{i=1}^{d} \frac{y_i^2}{v_i}. \tag{21}$$

Combining (18) and (21) proves the result. $\square$

A.2 Proof of Lemma 3

Lemma 3. *Let $p \in \Delta(\mathcal{X})$ be a weighted m-set distribution with positive weights and $1 \le m \le d-1$. Then its second-moment matrix M is positive definite and, for every $y \in H_0$,*

$$y^\top M^{-1}y = y^\top \Sigma^+ y.$$

Consequently, for every $x \in \mathcal{X}$,

$$x^\top M^{-1}x = 1 + (x-\mu)^\top \Sigma^+ (x-\mu),$$

where $\mu = \mu(p)$ is the marginal vector of p.

Proof. We first show that M is positive definite. Suppose that $a^\top Ma = 0$. Since

$$a^\top Ma = \mathbb{E}_{X\sim p}\big[(a^\top X)^2\big],$$

we have $a^\top x = 0$ for every $x \in \mathcal{X}$, because p has full support.

Fix $i \neq j$. Since $1 \le m \le d-1$, we can choose two actions that differ only by exchanging item i with item j. Comparing their inner products with a gives $a_i = a_j$. Thus, $a = c\mathbf{1}$ for some $c \in \mathbb{R}$. Since every $x \in \mathcal{X}$ has m nonzero coordinates,

$$0 = a^\top x = cm.$$

Therefore, $c = 0$, and hence $a = 0$. This proves that $M \succ 0$. By Lemma 2, $\ker(\Sigma) = \operatorname{span}\{\mathbf{1}\}$. Since Σ is symmetric, this also gives $\operatorname{range}(\Sigma) = H_0$. Every $X \in \mathcal{X}$ satisfies $\langle X, \mathbf{1}\rangle = m$. Therefore,

$$M\mathbf{1} = \mathbb{E}_{X\sim p}\big[X\langle X, \mathbf{1}\rangle\big] = m\mu.$$

Since M is invertible,

$$M^{-1}\mu = \frac{1}{m}\mathbf{1}. \tag{22}$$

Now fix $y \in H_0$ and let $z := M^{-1}y$. By (22),

$$\mu^\top z = (M^{-1}\mu)^\top y = \frac{1}{m}\mathbf{1}^\top y = 0.$$

It follows that

$$\Sigma z = (M - \mu\mu^\top)z = y.$$

Moreover, $\Sigma^+ y$ also solves the equation $\Sigma u = y$. Therefore,

$$z - \Sigma^+ y \in \ker(\Sigma) = \operatorname{span}\{\mathbf{1}\}.$$

Since $y \in H_0$, it is orthogonal to this difference. Hence,

$$y^\top M^{-1} y = y^\top \Sigma^+ y. \tag{23}$$

Finally, fix $x \in \mathcal{X}$ and let $y := x - \mu$. Since $\mathbf{1}^\top x = \mathbf{1}^\top \mu = m$, we have $y \in H_0$. Expanding $x = \mu + y$ gives

$$\begin{aligned} x^\top M^{-1} x &= \mu^\top M^{-1} \mu + 2y^\top M^{-1} \mu + y^\top M^{-1} y \\ &= 1 + y^\top \Sigma^+ y, \end{aligned}$$

where the second equality follows from (22) and (23). Substituting $y = x - \mu$ proves that

$$x^\top M^{-1} x = 1 + (x - \mu)^\top \Sigma^+ (x - \mu).$$

This concludes the proof. □

A.3 Proof of Lemma 4

Lemma 4. *Let $p \in \mathcal{P}_\lambda$ be a weighted m-set distribution, where $1 \le m \le d-1$, and let μ and M be its marginal vector and second-moment matrix. For every $x \in \mathcal{X}$, define*

$$\phi_p(x) := 1 + 2 \sum_{i=1}^{d} \frac{(x_i - \mu_i)^2}{\mu_i(1-\mu_i)}.$$

Then, for every $x \in \mathcal{X}$, $x^\top M^{-1} x \le \phi_p(x)$. Moreover, $4\phi_p(x) = c_p^\top x$, where

$$c_{p,i} = 4\left(\frac{1 + 2\sum_{j=1}^{d} \frac{\mu_j}{1-\mu_j}}{m} + 2\frac{1 - 2\mu_i}{\mu_i(1-\mu_i)} \right) \qquad \text{for every } i \in [d].$$

Finally, $\mathbb{E}_{X\sim p}[\phi_p(X)] = 2d + 1$, and, for every $x \in \mathcal{X}$, $0 \le \phi_p(x) \le 1 + 4d/\lambda$.

Proof. For $x \in \mathcal{X}$, put $y = x - \mu \in H_0$. By Lemma 3,

$$x^\top M^{-1} x = 1 + y^\top \Sigma^+ y.$$

Lemma 2 gives

$$y^\top \Sigma^+ y \le 2 \sum_i \frac{y_i^2}{\mu_i(1-\mu_i)},$$

and hence $x^\top M^{-1} x \le \phi_p(x)$. Taking expectations and using $\mathbb{E}[(X_i - \mu_i)^2] = \mu_i(1-\mu_i)$ yields

$$\mathbb{E}_{X\sim p}[\phi_p(X)] = 2d + 1.$$

For the range bound, if $x_i = 1$, then

$$2\frac{(x_i - \mu_i)^2}{\mu_i(1-\mu_i)} = 2\frac{1-\mu_i}{\mu_i} \le \frac{2}{\lambda r};$$

if $x_i = 0$, then

$$2\frac{(x_i-\mu_i)^2}{\mu_i(1-\mu_i)} = 2\frac{\mu_i}{1-\mu_i} \le \frac{2}{\lambda(1-r)}.$$

Since x has m ones and $d-m$ zeros,

$$\phi_p(x) \le 1 + \frac{2m}{\lambda r} + \frac{2(d-m)}{\lambda(1-r)} = 1 + \frac{4d}{\lambda}.$$

Finally, because $x_i \in \{0,1\}$,

$$2\frac{(x_i-\mu_i)^2}{\mu_i(1-\mu_i)} = 2\frac{\mu_i}{1-\mu_i} + 2x_i\frac{1-2\mu_i}{\mu_i(1-\mu_i)}.$$

Thus

$$\phi_p(x) = 1 + 2\sum_i \frac{\mu_i}{1-\mu_i} + 2\sum_i x_i \frac{1-2\mu_i}{\mu_i(1-\mu_i)}.$$

Distribute the constant term evenly over the selected coordinates using $\sum_i x_i = m$, and multiply the resulting affine representation by four, to obtain $4\phi_p(x) = c_p^\top x$. □

B Proofs from Section 5.2

Assumption 1. *Throughout this appendix, $T \ge 1$ is an integer, $\delta \in (0,1)$,*

$$0 < \eta \le \frac{1}{64}, \qquad \lambda = 128\eta d \le \frac{1}{2}.$$

The loss sequence $(\ell_t)_{t=1}^T$ is predictable, that is, ℓ_t is $\mathcal{F}_{t-1}$-measurable for every $t \in [T]$, and satisfies the action-normalized condition in Section 2.

The initialization and the update in Algorithm 1 ensure that, at every round t, the sampling distribution p_t is a weighted m-set distribution with positive weights and $p_t \in \mathcal{P}_{\lambda/2}$. Hence, the assumptions of Lemmas 1 and 4 hold at every round.

Recall from Algorithm 1 that

$$\phi_t := \phi_{p_t}, \qquad c_t := c_{p_t}, \qquad c_t^\top x = 4\phi_t(x), \qquad Z_t(x) := \widehat{y}_t(x) - \eta c_t^\top x.$$

We also write

$$\widehat{y}_t(x) := \langle x, \widehat{\ell}_t \rangle = x^\top M_t^{-1} X_t Y_t.$$

Set

$$B := 1 + \frac{8d}{\lambda}.$$

Applying Lemma 4 with $\lambda/2$, for every $x \in \mathcal{X}$, we have

$$0 \le c_t^\top x \le 4B, \qquad \mathbb{E}_{X\sim p_t}[c_t^\top X] = 4(2d+1), \qquad x^\top M_t^{-1} x \le \frac{c_t^\top x}{4}.$$

Lemma 5. *Fix $t \in [T]$. Suppose that $0 < \eta \le 1/64$, $\lambda = 128\eta d \le 1/2$, the loss at round t satisfies the action-normalized condition, and p_t is a weighted m-set distribution with positive weights in $\mathcal{P}_{\lambda/2}$. Then, for every $x \in \mathcal{X}$, $\eta|Z_t(x)| \le 1/4$.*

Proof. Since $|Y_t| \le 1$, the Cauchy–Schwarz inequality in the norm induced by M_t^{-1} gives

$$|\widehat{y}_t(x)| = |x^\top M_t^{-1} X_t Y_t| \le \sqrt{x^\top M_t^{-1} x}\sqrt{X_t^\top M_t^{-1} X_t}.$$

Applying Lemma 4 to $p_t \in \mathcal{P}_{\lambda/2}$, both leverage scores in the last inequality are at most $\phi_t(x)$ and $\phi_t(X_t)$, respectively, and both ϕ_t-values are at most

$$B = 1 + \frac{8d}{\lambda}.$$

Thus

$$|\widehat{y}_t(x)| \le B.$$

Also $0 \le c_t^\top x \le 4B$. Hence

$$|Z_t(x)| = |\widehat{y}_t(x) - \eta c_t^\top x| \le B + 4\eta B.$$

The parameter choice implies

$$\eta B = \eta + \frac{8\eta d}{\lambda} = \eta + \frac{1}{16} \le \frac{5}{64}.$$

Therefore

$$\eta|Z_t(x)| \le (1+4\eta)\eta B \le \left(1 + \frac{1}{16}\right)\frac{5}{64} = \frac{85}{1024} < \frac{1}{4},$$

where we used $\eta \le 1/64$. This proves the claim. ☐

Lemma 6. *Under Assumption 1, for every distribution $q \in \mathcal{P}_\lambda$,*

$$\sum_{t=1}^{T} \left(\mathbb{E}_{X\sim p_t}[Z_t(X)] - \mathbb{E}_{X\sim q}[Z_t(X)]\right) \le \frac{\mathrm{KL}(q\|p_1)}{\eta} + 1 + 2\eta\sum_{t=1}^{T}\mathbb{E}_{X\sim p_t}[Z_t(X)^2].$$

Proof. Fix $t \in [T]$ and define the unprojected multiplicative-weights distribution

$$\widetilde{p}_{t+1}(x) := \frac{p_t(x)\exp(-\eta Z_t(x))}{\mathbb{E}_{X\sim p_t}[\exp(-\eta Z_t(X))]}, \qquad x \in \mathcal{X}.$$

Since $Z_t(x) = z_t^\top x$ and $p_t = p_{\theta_t}$, the distribution above is precisely $p_{\widetilde{\theta}_{t+1}}$ from Algorithm 1. The guarantee in Algorithm 1 therefore gives, for every $q \in \mathcal{P}_\lambda$,

$$\mathrm{KL}(q\|p_{t+1}) \le \mathrm{KL}(q\|\widetilde{p}_{t+1}) + \varepsilon_{\mathrm{p}}, \tag{24}$$

where we dropped the nonnegative term $\mathrm{KL}(p_{t+1}\|\widetilde{p}_{t+1})$. By the definition of $\widetilde{p}_{t+1}$,

$$\mathrm{KL}(q\|\widetilde{p}_{t+1}) = \mathrm{KL}(q\|p_t) + \eta\mathbb{E}_{X\sim q}[Z_t(X)] + \log\mathbb{E}_{X\sim p_t}[\exp(-\eta Z_t(X))].$$

Combining this identity with (24) and rearranging gives

$$\begin{aligned}\eta\left(\mathbb{E}_{X\sim p_t}[Z_t(X)] - \mathbb{E}_{X\sim q}[Z_t(X)]\right) \le{} & \mathrm{KL}(q\|p_t) - \mathrm{KL}(q\|p_{t+1}) \\ & + \eta\mathbb{E}_{X\sim p_t}[Z_t(X)] + \log\mathbb{E}_{X\sim p_t}[\exp(-\eta Z_t(X))] + \varepsilon_{\mathrm{p}}.\end{aligned}$$

By Lemma 5,

$$|\eta Z_t(x)| \le \frac{1}{4} \qquad \text{for every } x \in \mathcal{X}.$$

Using

$$e^{-u} \le 1 - u + 2u^2 \qquad \text{for } |u| \le \frac{1}{4},$$

we obtain

$$\mathbb{E}_{X\sim p_t}[\exp(-\eta Z_t(X))] \le 1 - \eta \mathbb{E}_{X\sim p_t}[Z_t(X)] + 2\eta^2 \mathbb{E}_{X\sim p_t}[Z_t(X)^2].$$

The inequality $\log(1+s) \le s$ then gives

$$\eta \mathbb{E}_{X\sim p_t}[Z_t(X)] + \log \mathbb{E}_{X\sim p_t}[\exp(-\eta Z_t(X))] \le 2\eta^2 \mathbb{E}_{X\sim p_t}[Z_t(X)^2].$$

Therefore,

$$\mathbb{E}_{X\sim p_t}[Z_t(X)] - \mathbb{E}_{X\sim q}[Z_t(X)] \le \frac{\mathrm{KL}(q\|p_t) - \mathrm{KL}(q\|p_{t+1})}{\eta} + 2\eta \mathbb{E}_{X\sim p_t}[Z_t(X)^2] + \frac{\varepsilon_{\mathrm{p}}}{\eta}.$$

Summing over t telescopes the KL terms. Finally, $\mathrm{KL}(q\|p_{T+1}) \ge 0$, so

$$\sum_{t=1}^{T} \left(\mathbb{E}_{X\sim p_t}[Z_t(X)] - \mathbb{E}_{X\sim q}[Z_t(X)]\right) \le \frac{\mathrm{KL}(q\|p_1)}{\eta} + 2\eta \sum_{t=1}^{T} \mathbb{E}_{X\sim p_t}[Z_t(X)^2] + \frac{T\varepsilon_{\mathrm{p}}}{\eta}.$$

Since $\varepsilon_{\mathrm{p}} = \eta/T$, the last term equals one. ∎

Lemma 7. *Under Assumption 1, let $\rho \in (0,1)$, and let $(q_t)_{t=1}^T$ be a predictable sequence of distributions in $\Delta(\mathcal{X})$. For every $t \in [T]$, define*

$$A_t(q_t) := \mathbb{E}_{X\sim q_t}[\langle X, \ell_t\rangle], \qquad \widehat{A}_t(q_t) := \mathbb{E}_{X\sim q_t}[\widehat{y}_t(X)].$$

Then, with probability at least $1-\rho$,

$$\left| \sum_{t=1}^{T} \left(\widehat{A}_t(q_t) - A_t(q_t) \right) \right| \le \frac{\eta}{4} \sum_{t=1}^{T} \mathbb{E}_{X\sim q_t}[c_t^\top X] + \frac{\log(2/\rho)}{\eta}.$$

Proof. For every $t \in [T]$, define

$$\Delta_t := \widehat{A}_t(q_t) - A_t(q_t).$$

Recall that $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot \mid \mathcal{F}_{t-1}]$. Since p_t, q_t, and ℓ_t are predictable, they are fixed conditionally on $\mathcal{F}_{t-1}$. The only randomness at round t comes from $X_t \sim p_t$. We first show that Δ_t has conditional mean zero. By the definition of the KW loss estimate,

$$\mathbb{E}_t[\widehat{\ell}_t] = M_t^{-1} \mathbb{E}_t[X_t X_t^\top] \ell_t = M_t^{-1} M_t \ell_t = \ell_t.$$

Therefore,

$$\mathbb{E}_t[\widehat{A}_t(q_t)] = A_t(q_t), \qquad \mathbb{E}_t[\Delta_t] = 0.$$

Define $\bar{x}_t := \mathbb{E}_{X\sim q_t}[X]$. Then

$$\widehat{A}_t(q_t) = \bar{x}_t^\top M_t^{-1} X_t Y_t.$$

Since $|Y_t| \le 1$,

$$\mathbb{E}_t[\widehat{A}_t(q_t)^2] \le \bar{x}_t^\top M_t^{-1} \mathbb{E}_t[X_t X_t^\top] M_t^{-1} \bar{x}_t = \bar{x}_t^\top M_t^{-1} \bar{x}_t.$$

The matrix M_t^{-1} is positive semidefinite. Hence, Jensen's inequality gives

$$\bar{x}_t^\top M_t^{-1} \bar{x}_t \le \mathbb{E}_{X\sim q_t}\left[X^\top M_t^{-1} X\right].$$

By Lemma 4 and $c_t^\top X = 4\phi_t(X)$,

$$X^\top M_t^{-1} X \le \frac{c_t^\top X}{4}.$$

It follows that

$$\mathbb{E}_t[\widehat{A}_t(q_t)^2] \le \frac{1}{4}\mathbb{E}_{X\sim q_t}[c_t^\top X].$$

Since $\mathbb{E}_t[\widehat{A}_t(q_t)] = A_t(q_t)$, we obtain

$$\mathbb{E}_t[\Delta_t^2] \le \frac{1}{4}\mathbb{E}_{X\sim q_t}[c_t^\top X]. \tag{25}$$

We next bound Δ_t uniformly. Since $c_t^\top x \le 4B$ for every $x \in \mathcal{X}$, the previous leverage bound gives

$$\bar{x}_t^\top M_t^{-1}\bar{x}_t \le B, \qquad X_t^\top M_t^{-1} X_t \le B.$$

Using Cauchy–Schwarz in the norm induced by M_t^{-1}, we obtain

$$|\widehat{A}_t(q_t)| = \left|\bar{x}_t^\top M_t^{-1} X_t Y_t\right| \le \sqrt{\bar{x}_t^\top M_t^{-1}\bar{x}_t}\sqrt{X_t^\top M_t^{-1} X_t} \le B.$$

The action-normalized condition gives $|A_t(q_t)| \le 1$. Since $B \ge 1$,

$$|\Delta_t| \le B + 1 \le 2B.$$

Moreover,

$$\eta B = \eta + \frac{8\eta d}{\lambda} = \eta + \frac{1}{16} \le \frac{5}{64}.$$

Therefore,

$$|\eta\Delta_t| \le 2\eta B \le \frac{5}{32} < 1. \tag{26}$$

For every u with $|u| \le 1$,

$$e^u \le 1 + u + u^2.$$

Using (26), $\mathbb{E}_t[\Delta_t] = 0$, and (25), we obtain

$$\begin{aligned}
\mathbb{E}_t[\exp(\eta\Delta_t)] &\le 1 + \eta^2\mathbb{E}_t[\Delta_t^2] \\
&\le \exp\left(\eta^2\mathbb{E}_t[\Delta_t^2]\right) \\
&\le \exp\left(\frac{\eta^2}{4}\mathbb{E}_{X\sim q_t}[c_t^\top X]\right).
\end{aligned}$$

For $s \in \{0, \dots, T\}$, define

$$\mathcal{M}_s := \exp\left(\eta\sum_{t=1}^{s}\Delta_t - \frac{\eta^2}{4}\sum_{t=1}^{s}\mathbb{E}_{X\sim q_t}[c_t^\top X]\right).$$

The previous conditional moment bound shows that $(\mathcal{M}_s)_{s=0}^T$ is a nonnegative supermartingale with $\mathcal{M}_0 = 1$. Hence, $\mathbb{E}[\mathcal{M}_T] \le 1$. By Markov's inequality, with probability at least $1 - \rho/2$,

$$\sum_{t=1}^{T}\Delta_t \le \frac{\eta}{4}\sum_{t=1}^{T}\mathbb{E}_{X\sim q_t}[c_t^\top X] + \frac{\log(2/\rho)}{\eta}.$$

The same argument applied to $-\Delta_t$ gives, with probability at least $1-\rho/2$,

$$-\sum_{t=1}^{T}\Delta_t \le \frac{\eta}{4}\sum_{t=1}^{T}\mathbb{E}_{X\sim q_t}[c_t^\top X] + \frac{\log(2/\rho)}{\eta}.$$

A union bound over the two events proves the result. □

Lemma 8. *Under Assumption 1, for every $\rho \in (0,1)$, with probability at least $1-\rho$,*

$$\sum_{t=1}^{T}\phi_t(X_t) \le 2(2d+1)T + 2\left(1+\frac{8d}{\lambda}\right)\log\frac{1}{\rho}.$$

Proof. Recall that $B = 1 + 8d/\lambda$. Conditionally on $\mathcal{F}_{t-1}$, the distribution p_t and the function ϕ_t are fixed, while $X_t \sim p_t$. Therefore, Lemma 4 gives

$$\mathbb{E}_t[\phi_t(X_t)] = \mathbb{E}_{X\sim p_t}[\phi_t(X)] = 2d+1, \qquad 0 \le \phi_t(X_t) \le B.$$

Since $\phi_t(X_t)/B \in [0,1]$, convexity of the exponential gives

$$e^u \le 1 + (e-1)u, \qquad u \in [0,1].$$

It follows that

$$\mathbb{E}_t\left[\exp\left(\frac{\phi_t(X_t)}{B}\right)\right] \le 1 + \frac{(e-1)(2d+1)}{B} \le \exp\left(\frac{(e-1)(2d+1)}{B}\right).$$

For $s \in \{0,\dots,T\}$, define

$$\mathcal{M}_s := \exp\left(\frac{1}{B}\sum_{t=1}^{s}\phi_t(X_t) - \frac{(e-1)(2d+1)s}{B}\right).$$

The previous conditional moment bound shows that $(\mathcal{M}_s)_{s=0}^T$ is a nonnegative supermartingale with $\mathcal{M}_0 = 1$. Hence,

$$\mathbb{E}[\mathcal{M}_T] \le 1.$$

By Markov's inequality, with probability at least $1-\rho$,

$$\sum_{t=1}^{T}\phi_t(X_t) \le (e-1)(2d+1)T + B\log\frac{1}{\rho}.$$

Using $e-1 \le 2$ and substituting the definition of B proves the result. □

Lemma 9. *Under Assumption 1, for every $\rho \in (0,1)$, with probability at least $1-\rho$,*

$$\sum_{t=1}^{T}\left(\langle X_t, \ell_t\rangle - \mathbb{E}_{X\sim p_t}[\langle X, \ell_t\rangle]\right) \le \sqrt{2T\log\frac{1}{\rho}}.$$

Proof. Define

$$D_t := \langle X_t, \ell_t\rangle - \mathbb{E}_{X\sim p_t}[\langle X, \ell_t\rangle].$$

Since p_t and ℓ_t are determined before X_t is sampled,

$$\mathbb{E}[D_t \mid \mathcal{F}_{t-1}] = 0.$$

Thus, $(D_t)_{t=1}^T$ is a martingale difference sequence.

By the action-normalized condition, $\langle X_t, \ell_t \rangle \in [-1, 1]$. Therefore, conditionally on $\mathcal{F}_{t-1}$, D_t belongs to an interval of length at most 2. The Azuma–Hoeffding inequality then gives, for every $u > 0$,

$$\mathbb{P}\left(\sum_{t=1}^T D_t \geq u\right) \leq \exp\left(-\frac{u^2}{2T}\right).$$

Taking $u = \sqrt{2T \log(1/\rho)}$ proves the result. $\square$

B.1 Proof of the main regret theorem

Theorem 1 (High-probability KW bound)**.** *There exists a universal constant $C > 0$ such that the following holds. Let $T \geq 1$ be an integer, let $\delta \in (0, 1)$, and assume $1 \leq m \leq d - 1$. Run Algorithm 1. Then, with probability at least $1 - \delta$,*

$$R_T \leq C\sqrt{dT\left(\log K + \log \frac{1}{\delta}\right)},$$

where $K = \binom{d}{m}$. One may take $C = 160$.

Proof. For every $x \in \mathcal{X}$, define the smoothed comparator

$$q_x := (1 - \lambda)\delta_x + \lambda U,$$

where δ_x assigns probability one to x, and U is the uniform distribution over $\mathcal{X}$. If $x_i = 1$, the i-th marginal of q_x is

$$1 - \lambda(1 - r),$$

while, if $x_i = 0$, it is λr. Therefore, $q_x \in \mathcal{P}_\lambda$. Moreover, since $p_1 = U$,

$$\mathrm{KL}(q_x \| p_1) = \mathrm{KL}(q_x \| U) = \log K - H(q_x) \leq \log K.$$

Applying Lemma 6 with $q = q_x$ gives

$$\sum_{t=1}^T \left(\mathbb{E}_{X\sim p_t}[Z_t(X)] - \mathbb{E}_{X\sim q_x}[Z_t(X)]\right) \leq \frac{\log K}{\eta} + 1 + 2\eta \sum_{t=1}^T \mathbb{E}_{X\sim p_t}[Z_t(X)^2]. \tag{27}$$

We first bound the second-order term. Since $Z_t(X) = \widehat{y}_t(X) - \eta c_t^\top X$, we have

$$Z_t(X)^2 \leq 2\widehat{y}_t(X)^2 + 2\eta^2 (c_t^\top X)^2.$$

Furthermore, we have

$$\begin{aligned}
\mathbb{E}_{X\sim p_t}[\widehat{y}_t(X)^2] &= Y_t^2 X_t^\top M_t^{-1} \mathbb{E}_{X\sim p_t}[XX^\top] M_t^{-1} X_t \\
&= Y_t^2 X_t^\top M_t^{-1} X_t \\
&\leq \phi_t(X_t),
\end{aligned}$$

where the last inequality follows from $|Y_t| \leq 1$ and Lemma 4. Recall that

$$0 \leq c_t^\top X \leq 4B, \qquad \mathbb{E}_{X\sim p_t}[c_t^\top X] = 4(2d + 1).$$

Hence,

$$\mathbb{E}_{X\sim p_t}[(c_t^\top X)^2] \le 4B\mathbb{E}_{X\sim p_t}[c_t^\top X] = 16B(2d+1).$$

It follows that

$$2\eta\mathbb{E}_{X\sim p_t}[Z_t(X)^2] \le 4\eta\phi_t(X_t) + 64\eta^3 B(2d+1). \tag{28}$$

Using $Z_t(X) = \widehat{y}_t(X) - \eta c_t^\top X$ in (27), and then applying (28), gives

$$\begin{aligned}&\sum_{t=1}^T \left(\mathbb{E}_{X\sim p_t}[\widehat{y}_t(X)] - \mathbb{E}_{X\sim q_x}[\widehat{y}_t(X)]\right)\\ &\qquad \le \frac{\log K}{\eta} + 1 + 4\eta\sum_{t=1}^T \phi_t(X_t) + 64\eta^3 B(2d+1)T + 4\eta(2d+1)T - \eta\sum_{t=1}^T \mathbb{E}_{X\sim q_x}[c_t^\top X]. \end{aligned} \tag{29}$$

We now apply Lemma 8 with $\rho = \delta/4$. With probability at least $1-\delta/4$,

$$4\eta\sum_{t=1}^T \phi_t(X_t) \le 8\eta(2d+1)T + 8\eta B\log\frac{4}{\delta}.$$

Since

$$\eta B = \eta + \frac{8\eta d}{\lambda} = \eta + \frac{1}{16} \le \frac{5}{64},$$

we have $8\eta B \le 5/8$. Therefore,

$$4\eta\sum_{t=1}^T \phi_t(X_t) \le 8\eta(2d+1)T + \log\frac{4}{\delta}. \tag{30}$$

Next, apply Lemma 7 to the predictable sequence $q_t = p_t$, with $\rho = \delta/4$. With probability at least $1-\delta/4$,

$$\sum_{t=1}^T \left(\mathbb{E}_{X\sim p_t}[\langle X, \ell_t\rangle] - \mathbb{E}_{X\sim p_t}[\widehat{y}_t(X)]\right) \le \eta(2d+1)T + \frac{\log(8/\delta)}{\eta}. \tag{31}$$

For every $x \in \mathcal{X}$, apply the same lemma to the constant predictable sequence $q_t = q_x$, with $\rho = \delta/(4K)$. A union bound over the K actions shows that, with probability at least $1-\delta/4$, simultaneously for every $x \in \mathcal{X}$,

$$\sum_{t=1}^T \left(\mathbb{E}_{X\sim q_x}[\widehat{y}_t(X)] - \mathbb{E}_{X\sim q_x}[\langle X, \ell_t\rangle]\right) \le \frac{\eta}{4}\sum_{t=1}^T \mathbb{E}_{X\sim q_x}[c_t^\top X] + \frac{\log(8K/\delta)}{\eta}. \tag{32}$$

Combining (29), (30), (31), and (32), we obtain, simultaneously for every $x \in \mathcal{X}$,

$$\begin{aligned}\sum_{t=1}^T \left(\mathbb{E}_{X\sim p_t}[\langle X, \ell_t\rangle] - \mathbb{E}_{X\sim q_x}[\langle X, \ell_t\rangle]\right) &\le \frac{\log K + \log(8/\delta) + \log(8K/\delta)}{\eta}\\ &\quad + 1 + 13\eta(2d+1)T + 64\eta^3 B(2d+1)T + \log\frac{4}{\delta} - \eta\sum_{t=1}^T \mathbb{E}_{X\sim q_x}[c_t^\top X] + \frac{\eta}{4}\sum_{t=1}^T \mathbb{E}_{X\sim q_x}[c_t^\top X]. \end{aligned} \tag{33}$$

The last two terms satisfy

$$-\eta \sum_{t=1}^{T} \mathbb{E}_{X \sim q_x}[c_t^\top X] + \frac{\eta}{4} \sum_{t=1}^{T} \mathbb{E}_{X \sim q_x}[c_t^\top X] = -\frac{3\eta}{4} \sum_{t=1}^{T} \mathbb{E}_{X \sim q_x}[c_t^\top X] \le 0,$$

because $c_t^\top X \ge 0$. Moreover, $64\eta^3 B = 64\eta^2(\eta B) \le 5\eta^2 \le 5\eta$. Therefore, (33) gives

$$\sum_{t=1}^{T} \left(\mathbb{E}_{X \sim p_t}[\langle X, \ell_t \rangle] - \mathbb{E}_{X \sim q_x}[\langle X, \ell_t \rangle]\right) \le \frac{\log K + \log(8/\delta) + \log(8K/\delta)}{\eta} + 1 + 18\eta(2d+1)T + \log\frac{4}{\delta}.$$

Using $2d + 1 \le 3d$ and enlarging the logarithmic terms, we obtain

$$\sum_{t=1}^{T} \left(\mathbb{E}_{X \sim p_t}[\langle X, \ell_t \rangle] - \mathbb{E}_{X \sim q_x}[\langle X, \ell_t \rangle]\right) \le \frac{3\log K + 3\log(12/\delta)}{\eta} + 1 + 54\eta dT + \log\frac{4}{\delta}. \tag{34}$$

Let $x^\star$ be a best action in hindsight. Since $q_{x^\star} = (1-\lambda)\delta_{x^\star} + \lambda U$ and every action loss belongs to $[-1, 1]$,

$$\mathbb{E}_{X \sim q_{x^\star}}[\langle X, \ell_t \rangle] \le \langle x^\star, \ell_t \rangle + 2\lambda.$$

Applying (34) with $x = x^\star$ gives

$$\begin{aligned}
&\sum_{t=1}^{T} \left(\mathbb{E}_{X \sim p_t}[\langle X, \ell_t \rangle] - \langle x^\star, \ell_t \rangle\right) \\
&\quad \le \frac{3\log K + 3\log(12/\delta)}{\eta} + 1 + 54\eta dT + 2\lambda T + \log\frac{4}{\delta} \\
&\quad \le \frac{3\log K + 3\log(12/\delta)}{\eta} + 1 + 310\eta dT + \log\frac{4}{\delta},
\end{aligned}$$

where the last inequality uses $\lambda = 128\eta d$. Finally, apply Lemma 9 with $\rho = \delta/4$. With probability at least $1 - \delta/4$,

$$\sum_{t=1}^{T} \left(\langle X_t, \ell_t \rangle - \mathbb{E}_{X \sim p_t}[\langle X, \ell_t \rangle]\right) \le \sqrt{2T \log\frac{4}{\delta}} \le 2\sqrt{T \log\frac{4}{\delta}}.$$

Since $310 \le 320$ and $1 + \log(4/\delta) \le 4\log(12/\delta)$, a union bound over the four failure events gives

$$R_T \le \frac{3\log K + 3\log(12/\delta)}{\eta} + 320\eta dT + 4\log\frac{12}{\delta} + 2\sqrt{T\log\frac{12}{\delta}}. \tag{35}$$

It remains to substitute the value of η. Define

$$A := \log K + \log\frac{12}{\delta}.$$

Also set

$$S := \log K + \log\frac{1}{\delta}.$$

Since $K = \binom{d}{m} \ge 2$, we have $S \ge \log 2$, and therefore

$$A = S + \log 12 \le \left(1 + \frac{\log 12}{\log 2}\right) S \le \frac{23}{5} S, \tag{36}$$

where the last inequality follows from $12^5 < 2^{18}$. If $\eta = \sqrt{A/(320dT)}$, then the first two terms in (35) are exactly

$$\frac{3A}{\eta} + 320\eta dT = 4\sqrt{320dTA}.$$

Moreover,

$$4\log\frac{12}{\delta} \le 4A, \qquad 2\sqrt{T\log\frac{12}{\delta}} \le 2\sqrt{dTA}.$$

It remains to consider the case in which $\eta = 1/(256d)$. In this case,

$$\sqrt{\frac{A}{320dT}} \ge \frac{1}{256d},$$

which implies

$$T \le \frac{256^2}{320}\, dA.$$

Since the action-normalized condition gives the deterministic bound $R_T \le 2T$, we also have

$$R_T \le 2\sqrt{\frac{256^2}{320}}\sqrt{dTA} \le 29\sqrt{dTA}.$$

In the non-clipped case, Equation (35) is at most $74\sqrt{dTA} + 4A$. Since $29 < 74$ and $A > 0$, the same bound also holds in the clipped case. Hence, by (36),

$$R_T \le 74\sqrt{\frac{23}{5}}\sqrt{dTS} + \frac{92}{5}S. \tag{37}$$

Set

$$\gamma := \sqrt{\frac{S}{dT}} > 0.$$

Since

$$S = \gamma\sqrt{dTS},$$

if $\gamma \le 1/160$, then

$$\begin{aligned} R_T &\le \left(74\sqrt{\frac{23}{5}} + \frac{92}{5}\gamma\right)\sqrt{dTS} \\ &\le \left(74\frac{43}{20} + \frac{92}{5}\frac{1}{160}\right)\sqrt{dTS} \\ &= \frac{31843}{200}\sqrt{dTS} < 160\sqrt{dTS}, \end{aligned}$$

where we used

$$\sqrt{\frac{23}{5}} < \frac{43}{20}.$$

If instead $\gamma > 1/160$, the deterministic bound $R_T \le 2T$ and $d \ge 2$ give

$$R_T \le 2T = \frac{2}{d\gamma}\sqrt{dTS} \le \frac{1}{\gamma}\sqrt{dTS} < 160\sqrt{dTS}.$$

Thus, in all cases,

$$R_T \le 160\sqrt{dT\left(\log K + \log\frac{1}{\delta}\right)},$$

which proves the theorem with $C = 160$. $\square$

C Polynomial-time implementation

This appendix proves Proposition 1. We show that our algorithm can be implemented in polynomial time without enumerating $\mathcal{X}$. It maintains each p_t as a weighted m-set distribution, computes the required moments and draws samples in polynomial time, and reduces the target KL projection to a convex problem in $2d$ variables. The convex problem is solved only to the inverse-polynomial accuracy specified below.

C.1 Updates and KL projection

We first identify the exact KL projection toward which the algorithm computes and show that it reduces to a convex problem in $2d$ variables. We then prove that solving this problem to an explicit objective accuracy is sufficient for the approximate update in Algorithm 1.

Suppose that $p_t = p_{\theta_t}$. Define the unprojected multiplicative-weights update

$$\widetilde{p}_{t+1}(x) := \frac{p_t(x)\exp(-\eta Z_t(x))}{\mathbb{E}_{X\sim p_t}[\exp(-\eta Z_t(X))]}.$$

For every $p \in \mathcal{P}_\lambda$,

$$\eta\mathbb{E}_{X\sim p}[Z_t(X)] + \mathrm{KL}(p\|p_t) = \mathrm{KL}(p\|\widetilde{p}_{t+1}) - \log \mathbb{E}_{X\sim p_t}[\exp(-\eta Z_t(X))].$$

The last term does not depend on p. Therefore, the exact constrained OMD iterate is the KL projection of $\widetilde{p}_{t+1}$ onto $\mathcal{P}_\lambda$, namely,

$$p^\star_{t+1} \in \arg\min_{p\in\mathcal{P}_\lambda} \mathrm{KL}(p\|\widetilde{p}_{t+1}).$$

The algorithm computes the approximate replacement specified in Algorithm 1 rather than assuming access to this exact projection.

We next show that the first step preserves the weighted m-set form. Since $c_t^\top x$ is affine in x, the surrogate loss is affine on $\mathcal{X}$:

$$Z_t(x) = \langle x, \widehat{\ell}_t\rangle - \eta c_t^\top x = z_t^\top x, \qquad z_t := \widehat{\ell}_t - \eta c_t.$$

Consequently,

$$\widetilde{p}_{t+1}(x) = \frac{p_{\theta_t}(x)\exp(-\eta z_t^\top x)}{\mathbb{E}_{X\sim p_{\theta_t}}[\exp(-\eta z_t^\top X)]} = p_{\widetilde{\theta}_{t+1}}(x), \qquad \widetilde{\theta}_{t+1} := \theta_t - \eta z_t. \tag{38}$$

Hence, if p_t is a weighted m-set distribution, then so is $\widetilde{p}_{t+1}$.

It remains to analyze the target KL projection. The next lemma shows that the projection of a weighted m-set distribution onto $\mathcal{P}_\lambda$ is again a weighted m-set distribution. It also shows that the projection can be obtained by solving a convex problem in $2d$ variables.

Lemma 10. *Define*

$$a_i := \lambda r, \qquad b_i := 1 - \lambda(1-r), \qquad i \in [d].$$

For every $\widetilde{\theta} \in \mathbb{R}^d$, the KL projection

$$\arg\min_{p\in\mathcal{P}_\lambda} \mathrm{KL}(p\|p_{\widetilde{\theta}})$$

is a weighted m-set distribution p_{θ^+}, where $\theta^+ = \widetilde{\theta} + \alpha^\star - \beta^\star$. Here, $(\alpha^\star, \beta^\star) \in \mathbb{R}^d_+ \times \mathbb{R}^d_+$ minimizes the convex function

$$\Psi_{\widetilde{\theta}}(\alpha, \beta) := F(\widetilde{\theta} + \alpha - \beta) - \alpha^\top a + \beta^\top b.$$

Moreover,

$$\frac{\partial \Psi_{\widetilde{\theta}}}{\partial \alpha_i} = \mu_i(\widetilde{\theta} + \alpha - \beta) - a_i, \qquad \frac{\partial \Psi_{\widetilde{\theta}}}{\partial \beta_i} = b_i - \mu_i(\widetilde{\theta} + \alpha - \beta).$$

Proof. The projection minimizes $\mathrm{KL}(p \| p_{\widetilde{\theta}})$ over $p \in \Delta(\mathcal{X})$, subject to

$$a_i - \mathbb{E}_{X \sim p}[X_i] \le 0, \qquad \mathbb{E}_{X \sim p}[X_i] - b_i \le 0.$$

The uniform distribution has all marginals equal to r, strictly between a_i and b_i. Hence, Slater's condition holds. We may therefore use the standard strong-duality and KKT results of Boyd and Vandenberghe [6, Sections 5.2.3 and 5.5.3].

Introduce multipliers $\alpha_i, \beta_i \ge 0$. The Lagrangian is

$$\mathcal{L}(p, \alpha, \beta) = \mathrm{KL}(p \| p_{\widetilde{\theta}}) + \alpha^\top a - \beta^\top b + \mathbb{E}_{X \sim p}[(\beta - \alpha)^\top X].$$

For fixed (α, β), minimizing over p gives

$$p(x) \propto p_{\widetilde{\theta}}(x) e^{(\alpha - \beta)^\top x} = p_{\widetilde{\theta} + \alpha - \beta}(x).$$

The resulting dual maximization is equivalent to minimizing $\Psi_{\widetilde{\theta}}$. Strong duality proves the first claim, and the derivative identities follow from $\nabla F = \mu$. $\square$

Lemma 11. *Let $0 < \lambda \le 1/2$, let $\varepsilon > 0$, and let $\widetilde{\theta} \in \mathbb{R}^d$ satisfy*

$$\mathrm{KL}(U \| p_{\widetilde{\theta}}) \le T.$$

Define

$$h := \frac{\lambda}{2} \min\{r, 1 - r\} = \frac{\lambda s}{2d}, \qquad R := 2dT,$$

and

$$\tau := \min\left\{2h^2, \frac{\varepsilon}{2}, \frac{\varepsilon^2}{2R^2}\right\}. \tag{39}$$

Let

$$\mathcal{D}_R := \left\{(\alpha, \beta) \in \mathbb{R}^d_+ \times \mathbb{R}^d_+ : \|\alpha\|_1 + \|\beta\|_1 \le R\right\}.$$

Suppose that $(\alpha, \beta) \in \mathcal{D}_R$ satisfies

$$\Psi_{\widetilde{\theta}}(\alpha, \beta) \le \min_{(u,v) \in \mathcal{D}_R} \Psi_{\widetilde{\theta}}(u, v) + \tau. \tag{40}$$

Set

$$\theta^+ := \widetilde{\theta} + \alpha - \beta, \qquad p^+ := p_{\theta^+}.$$

Then $p^+ \in \mathcal{P}_{\lambda/2}$, and, for every $q \in \mathcal{P}_\lambda$,

$$\mathrm{KL}(q \| p^+) + \mathrm{KL}(p^+ \| p_{\widetilde{\theta}}) \le \mathrm{KL}(q \| p_{\widetilde{\theta}}) + \varepsilon. \tag{41}$$

Moreover, a pair satisfying (40) can be computed by the ellipsoid method in time

$$\mathrm{poly}\left(d, m, \log R, \log \frac{1}{\tau}\right).$$

Proof. Write

$$a_i := \lambda r, \qquad b_i := 1 - \lambda(1-r).$$

Let $(\alpha^\star, \beta^\star)$ be an exact minimizer of $\Psi_{\widetilde{\theta}}$, let

$$\theta^\star := \widetilde{\theta} + \alpha^\star - \beta^\star, \qquad p^\star := p_{\theta^\star},$$

and write $\mu^\star := \mu(p^\star)$. We first show that an exact minimizer belongs to $\mathcal{D}_R$. By strong duality, the value of the dual function at $(\alpha^\star, \beta^\star)$ equals the value of the KL projection problem and is therefore nonnegative. Evaluating the Lagrangian from the proof of Lemma 10 at the uniform distribution gives

$$0 \le \mathrm{KL}(U \| p_{\widetilde{\theta}}) - (1-\lambda) r \|\alpha^\star\|_1 - (1-\lambda)(1-r)\|\beta^\star\|_1.$$

Consequently,

$$\|\alpha^\star\|_1 + \|\beta^\star\|_1 \le \frac{T}{(1-\lambda)\min\{r, 1-r\}} \le \frac{2dT}{s} \le R.$$

Thus the minimum over $\mathcal{D}_R$ in (40) is the global minimum.

Let $p := p^+$, let $\mu := \mu(p)$, and abbreviate $\Psi := \Psi_{\widetilde{\theta}}$. The KKT conditions for the exact projection give

$$a_i \le \mu_i^\star \le b_i, \qquad \alpha_i^\star(\mu_i^\star - a_i) = 0, \qquad \beta_i^\star(b_i - \mu_i^\star) = 0.$$

Using

$$\mathrm{KL}(p^\star \| p) = F(\theta^+) - F(\theta^\star) - (\mu^\star)^\top (\theta^+ - \theta^\star),$$

and the complementary-slackness identities above, direct expansion gives the exact identity

$$\Psi(\alpha, \beta) - \Psi(\alpha^\star, \beta^\star) = \mathrm{KL}(p^\star \| p) + \alpha^\top(\mu^\star - a) + \beta^\top(b - \mu^\star). \tag{42}$$

All three terms on the right-hand side are nonnegative. Hence,

$$\mathrm{KL}(p^\star \| p) \le \tau.$$

Pinsker's inequality and the fact that every coordinate of an action belongs to $[0,1]$ imply

$$\|\mu - \mu^\star\|_\infty \le \mathrm{TV}(p, p^\star) \le \sqrt{\frac{\tau}{2}} \le h.$$

Every marginal of $p^\star$ belongs to $[\lambda r, 1-\lambda(1-r)]$. The distance from this interval to the corresponding boundary of $[\lambda r/2, 1 - \lambda(1-r)/2]$ is at least h. Therefore, $p \in \mathcal{P}_{\lambda/2}$.

Define the residual

$$\kappa := \alpha^\top(\mu - a) + \beta^\top(b - \mu).$$

By (42) and $(\alpha, \beta) \in \mathcal{D}_R$,

$$\begin{aligned} \kappa &= \alpha^\top(\mu^\star - a) + \beta^\top(b - \mu^\star) + (\alpha - \beta)^\top(\mu - \mu^\star) \\ &\le \tau + R\sqrt{\frac{\tau}{2}} \le \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \end{aligned}$$

where the last inequality follows from (39).

For every $q \in \mathcal{P}_\lambda$, the exponential-family representation gives the three-point identity

$$\begin{aligned} &\mathrm{KL}(q \| p_{\widetilde{\theta}}) - \mathrm{KL}(q \| p) - \mathrm{KL}(p \| p_{\widetilde{\theta}}) \\ &= (\mu(q) - \mu)^\top (\alpha - \beta). \end{aligned}$$

Since $a \le \mu(q) \le b$ coordinatewise and $\alpha, \beta \ge 0$, the right-hand side is at least $-\kappa$. The bound $\kappa \le \varepsilon$ proves (41).

It remains to justify the computational claim. Put

$$L := \sqrt{2d}, \qquad r_0 := \frac{R}{8d}, \qquad w_0 := \frac{R}{4d}\mathbf{1}_{2d}.$$

For $v \in \mathbb{R}^{2d}$ and $r > 0$, write $B(v, r) := \{u \in \mathbb{R}^{2d} : \|u - v\|_2 \le r\}$. The set $\mathcal{D}_R$ has an explicit separation procedure, is contained in the Euclidean ball of radius R, and contains $B(w_0, r_0)$. Indeed, every point of this ball has nonnegative coordinates and ℓ_1-norm at most $R/2 + \sqrt{2d}\, r_0 \le R$. By the derivative formulas in Lemma 10,

$$\|\nabla \Psi_{\tilde{\theta}}(w)\|_2 \le L \qquad \text{for every } w \in \mathbb{R}^{2d}.$$

Thus $\Psi_{\tilde{\theta}}$ is globally L-Lipschitz. Its value and gradient can be evaluated using $\mathrm{poly}(d, m)$ operations by Lemma 12.

Let $w^\star$ minimize $\Psi_{\tilde{\theta}}$ over $\mathcal{D}_R$. For $0 < \zeta \le r_0$, define

$$w_\zeta := \left(1 - \frac{\zeta}{r_0}\right) w^\star + \frac{\zeta}{r_0} w_0.$$

Convexity of $\mathcal{D}_R$ and $B(w_0, r_0) \subseteq \mathcal{D}_R$ imply $B(w_\zeta, \zeta) \subseteq \mathcal{D}_R$. Moreover, $\|w^\star - w_0\|_2 \le 2R$, and hence

$$\Psi_{\tilde{\theta}}(w_\zeta) \le \Psi_{\tilde{\theta}}(w^\star) + 16dL\zeta.$$

Set

$$\zeta := \min\left\{ r_0, \frac{\tau}{1 + (16d + 1)L} \right\}.$$

Weak constrained convex-function minimization returns a point $\widehat{w}$ at Euclidean distance at most ζ from $\mathcal{D}_R$, whose objective is at most ζ above the objective of every point whose closed ζ-ball is contained in $\mathcal{D}_R$ [15, Problem (2.1.22) and Theorem 4.3.13]. In particular,

$$\Psi_{\tilde{\theta}}(\widehat{w}) \le \Psi_{\tilde{\theta}}(w^\star) + (16dL + 1)\zeta.$$

Let $\overline{w}$ be the Euclidean projection of $\widehat{w}$ onto $\mathcal{D}_R$. Since $\mathcal{D}_R$ is the nonnegative ℓ_1-ball, its coordinates have the form

$$\overline{w}_i = \max\{\widehat{w}_i - \chi, 0\},$$

where $\chi = 0$ if the positive part of $\widehat{w}$ has ℓ_1-norm at most R, and otherwise χ is chosen so that $\sum_i \max\{\widehat{w}_i - \chi, 0\} = R$. Sorting the coordinates finds χ in $O(d \log d)$ time. Furthermore, $\|\overline{w} - \widehat{w}\|_2 \le \zeta$, so global Lipschitzness gives

$$\begin{aligned} \Psi_{\tilde{\theta}}(\overline{w}) &\le \Psi_{\tilde{\theta}}(w^\star) + \big(1 + (16d + 1)L\big)\zeta \\ &\le \Psi_{\tilde{\theta}}(w^\star) + \tau. \end{aligned}$$

Writing $\overline{w} = (\alpha, \beta)$ gives a pair in $\mathcal{D}_R$ satisfying (40). Since $R/r_0 = 8d$ and $\log(1/\zeta) = O(\log d + \log R + \log(1/\tau))$, the weak optimizer and the final projection use $\mathrm{poly}(d, m, \log R, \log(1/\tau))$ operations. $\square$

C.2 Moment and sampling routines

We now show how to compute the moments of a weighted m-set distribution and draw samples from it without enumerating $\mathcal{X}$. Given its parameters θ, we compute its marginals, its second-moment matrix, and an exact sample using dynamic programs for elementary symmetric polynomials.

Lemma 12. *Given $\theta \in \mathbb{R}^d$, the log-partition function $F(\theta)$, all marginals $\mu_i(p_\theta)$, all pairwise marginals*

$$\pi_{ij}(p_\theta) := \mathbb{P}_{X\sim p_\theta}(X_i = X_j = 1),$$

and a sample from p_θ can be drawn using a number of arithmetic operations polynomial in d and m, without enumerating $\mathcal{X}$.

Proof. Let $w_i := e^{\theta_i}$, and define

$$E[i,k] := e_k(w_1, \dots, w_i).$$

The recurrence

$$E[i,k] = E[i-1,k] + w_i E[i-1,k-1] \tag{43}$$

computes all entries with $0 \le i \le d$ and $0 \le k \le m$ in $O(dm)$ arithmetic operations. In particular, $E[0,0] = 1$ and $E[i,k] = 0$ whenever $k < 0$ or $k > i$. Moreover,

$$F(\theta) = \log E[d,m].$$

Write $E_k := e_k(w)$. For each $i \in [d]$, define

$$E_0^{(-i)} := 1, \qquad E_k^{(-i)} := E_k - w_i E_{k-1}^{(-i)}.$$

For $i \neq j$, define

$$E_0^{(-i,-j)} := 1, \qquad E_k^{(-i,-j)} := E_k^{(-i)} - w_j E_{k-1}^{(-i,-j)}.$$

We use the value zero for both arrays at negative indices. These recurrences give

$$\mu_i(p_\theta) = \frac{w_i E_{m-1}^{(-i)}}{E_m}, \qquad \pi_{ii}(p_\theta) = \mu_i(p_\theta), \qquad \pi_{ij}(p_\theta) = \frac{w_i w_j E_{m-2}^{(-i,-j)}}{E_m} \quad (i \neq j). \tag{44}$$

All pairwise marginals can be obtained in $O(d^2 m)$ arithmetic operations. The recurrence is Method 2 of Chen and Liu [11, Section 2].

For sampling, start from $(i,k) = (d,m)$. At state (i,k), include item i with probability

$$\frac{w_i E[i-1,k-1]}{E[i,k]}. \tag{45}$$

If item i is selected, continue from $(i-1,k-1)$; otherwise, continue from $(i-1,k)$. The recurrence (43) shows that (45) is the correct conditional probability. Thus, the procedure samples from p_θ using $O(dm)$ arithmetic operations after the table has been computed. It is the reverse-indexed conditional-Bernoulli procedure of Chen and Liu [11, Section 4, Procedure 3]; it is also equivalent to their direct backward-path construction (Procedure 5) and to the diagonal-L specialization of Kulesza and Taskar [24, Section 3.1 and Algorithm 2]. ∎

C.3 Proof of Proposition 1

Proposition 1. *Algorithm 1 can be implemented in* $T \cdot \mathrm{poly}(d, m, \log T)$ *time and* $\mathrm{poly}(d, m)$ *space, without enumerating* $\mathcal{X}$*. At every round,* p_t *is a weighted* m*-set distribution* p_{θ_t} *represented by the* d *parameters* $\theta_{t,1}, \ldots, \theta_{t,d}$.

Proof. Recall that

$$\varepsilon_{\mathrm{p}} = \frac{\eta}{T}.$$

We prove by induction that every p_t is a weighted m-set distribution in $\mathcal{P}_{\lambda/2}$ and that

$$\mathrm{KL}(U \| p_t) \leq (t-1)\left(\frac{1}{2} + \varepsilon_{\mathrm{p}}\right). \tag{46}$$

This holds at $t = 1$, because $\theta_1 = 0$ and $p_1 = p_{\theta_1} = U$.

Suppose that the claim holds at round t. Given θ_t, Lemma 12 computes the marginals and the second-moment matrix M_t in $O(d^2 m)$ operations. The same marginals determine ϕ_t and c_t in $O(d)$ operations. The sampling procedure in Lemma 12 draws X_t from p_t without enumerating $\mathcal{X}$: at each step, it draws one uniform random variable and compares it with the conditional probability in (45).

By Lemma 3, M_t is positive definite. Therefore, M_t^{-1} and $\widehat{\ell}_t$ can be computed in $O(d^3)$ operations. The surrogate loss is represented by the d-dimensional vector $z_t = \widehat{\ell}_t - \eta c_t$, and (38) gives $\widetilde{p}_{t+1} = p_{\widetilde{\theta}_{t+1}}$.

We next verify the hypothesis of Lemma 11. By the definition of the unprojected update,

$$\mathrm{KL}(U \| \widetilde{p}_{t+1}) = \mathrm{KL}(U \| p_t) + \eta \mathbb{E}_{X \sim U}[Z_t(X)] + \log \mathbb{E}_{X \sim p_t}[e^{-\eta Z_t(X)}].$$

Under the induction hypothesis, Lemma 5 gives $|\eta Z_t(x)| \leq 1/4$ for every $x \in \mathcal{X}$. In particular,

$$\eta \mathbb{E}_{X \sim U}[Z_t(X)] \leq \frac{1}{4}, \qquad \log \mathbb{E}_{X \sim p_t}[e^{-\eta Z_t(X)}] \leq \log(e^{1/4}) = \frac{1}{4}.$$

Hence,

$$\mathrm{KL}(U \| \widetilde{p}_{t+1}) \leq \mathrm{KL}(U \| p_t) + \frac{1}{2} \leq \frac{t}{2} + (t-1)\varepsilon_{\mathrm{p}} \leq T,$$

where the last inequality uses $t \leq T$ and $T\varepsilon_{\mathrm{p}} = \eta \leq 1/64$.

Apply Lemma 11 with $\varepsilon = \varepsilon_{\mathrm{p}}$. It computes $(\alpha_t, \beta_t) \in \mathcal{D}_{2dT}$ so that (40) holds with the value τ in (39). Set

$$\theta_{t+1} := \widetilde{\theta}_{t+1} + \alpha_t - \beta_t, \qquad p_{t+1} := p_{\theta_{t+1}}.$$

The lemma gives $p_{t+1} \in \mathcal{P}_{\lambda/2}$ and the KL inequality required in Algorithm 1. Since $U \in \mathcal{P}_\lambda$, that inequality also gives

$$\mathrm{KL}(U \| p_{t+1}) \leq \mathrm{KL}(U \| \widetilde{p}_{t+1}) + \varepsilon_{\mathrm{p}} \leq t\left(\frac{1}{2} + \varepsilon_{\mathrm{p}}\right).$$

This closes the induction.

It remains to bound the number of operations uniformly over the rounds. Since $K \geq 2$ and $\delta < 1$, the learning rate in Algorithm 1 satisfies

$$\eta \geq \frac{1}{256 d\sqrt{T}}.$$

Consequently,

$$\varepsilon_{\mathrm{p}} \geq \frac{1}{256dT^{3/2}}, \qquad \frac{\lambda s}{2d} \geq \frac{1}{4d\sqrt{T}}, \qquad R = 2dT.$$

The value τ in (39) therefore satisfies

$$\tau \geq \frac{1}{2^{19}d^4T^5}.$$

Thus $\log R + \log(1/\tau) = O(\log(dT))$, and Lemma 11 runs in $\mathrm{poly}(d, m, \log T)$ time per round. All other steps run in time polynomial in d and m. The algorithm therefore runs in $T \cdot \mathrm{poly}(d, m, \log T)$ time, uses $\mathrm{poly}(d, m)$ space, and never enumerates $\mathcal{X}$. $\square$

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